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Low emission zones and population health☆

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ABSTRACT

Air pollution has a detrimental impact on population health, but the effectiveness of policy measures targeting pollution is underexplored. I exploit the natural experiment generated by the staggered implementation of low emission zones in large cities across Germany to assess their impact on health. Using register data on outpatient and inpatient health care, I find that low emission zones reduce the number of patients with cardiovascular disease by 2–3%. This effect is particularly pronounced for those over the age of 65. The findings suggest that low emission zones can be an effective way to reduce air pollution and improve health.

1. Introduction

Traffic accounts for more than one quarter of ambient air pollution in urban areas (Umweltbundesamt, 2018). Direct user charges, congestion pricing, license plate based restrictions, and low emission zones have become popular tools to reduce traffic-induced pollution (Davis, 2008; Wolff, 2014; Simeonova et al., 2018). Policies that aim to improve air quality primarily target population health: pollutants such as particulate matter, nitrogen oxides, and ozone can exacerbate already existing medical conditions, and in some cases even cause them. Despite the considerable attention paid to the traffic and pollution effects of air quality policies, their effectiveness in improving population health remains underexplored.

This paper studies the effects of one particular regulatory instrument, namely low emission zones (LEZs), on air pollution and cardiovascular health. LEZs are designated areas that restrict cars' access based on their emission class. Since 2007, multiple cities in Europe and specifically in Germany have gradually been introducing such zones. Across countries and cities, LEZs vary in their operating hours and the restrictiveness of their vehicle exclusion. LEZs in Germany are among the most restrictive, as they impose a 24/7 ban on all days, and the restrictions apply to all vehicles, with few exemptions. The rollout started in 2008 after it became clear that most major cities were

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not in compliance with the EU Air Quality Standards. In Germany, LEZs were introduced in three phases, with each phase excluding an additional emission class of vehicles. The rich temporal and spatial variation in the implementation of LEZs combined with their restrictiveness generates a compelling natural experiment to evaluate their impact.

To study the effect of LEZs on air pollution and cardiovascular health, I exploit the variation across space and time in the introduction of LEZs in German cities using a difference-in-differences design. I use air pollution measurements from the German Environment Agency covering the period 2004–2016. I focus on two criteria pollutants: particulate matter with an aerodynamic diameter of less than 10 (PM_{10}) and nitrogen dioxide (NO_2), as vehicle exhaust is a dominant source of their emission.[1] I supplement the analysis with data from the Federal Motor Transport Authority for the years 2008–2017 to examine the impact of LEZs on the car fleet composition as a potential channel through which LEZs may reduce air pollution. To evaluate the impact on health, I use novel administrative outpatient care data from the universe of patients in the German statutory health insurance system during the period 2009–2017. I complement the outpatient data with evidence from a 70% random sample from German hospital diagnosis statistics for the years 2004–2014.

I find that LEZs reduce monthly PM_{10} concentrations by 2–3%. The reductions in monthly NO_2 concentrations are smaller in magnitude and often less precisely estimated. These effects corroborate the findings from previous evaluations of LEZs in terms of their effect on air pollution (Wolff, 2014; Malina and Scheffler, 2015; Gehrsitz, 2017). I further show that a likely channel through which LEZs affect air quality is by reducing the share of cars with the highest emissions in the treated cities.

By improving the air quality, LEZs also improve population health. Findings based on outpatient data suggest that the number of patients with cardiovascular disease decreases by 2–3%. Consistent with epidemiological studies, age and disease subgroup analysis shows that the effects are particularly strong for those above the age of 65 and for cerebrovascular diseases (7–12.6%). Event study plots show that changes in health outcomes occur roughly in parallel to changes in PM_{10}. Using complementary data from the Leibniz Institute for Economic Research (RWI), I further show that socioeconomic characteristics within LEZs do not change systematically over time compared to control areas, suggesting that estimates are not driven by compositional changes. Multiple robustness tests, such as the use of different control groups, sample restrictions, different definitions of outcome and treatment variables, and falsification checks suggest that the results are not sensitive to these alternative specifications. Findings from the hospital admission data also suggest a drop in the number of people admitted with cardiovascular disease, but with less precisely estimated point estimates.

Several epidemiological studies highlight the adverse association between air pollution, in particular PM_{10}, and cardiovascular health (Linn et al., 2000; Pope, 2000; Tsai et al., 2003; Brook et al., 2004; Wellenius et al., 2005; Franchini and Mannucci, 2007; Fiordelisi et al., 2017). This negative relationship can be observed even at levels below commonly applied concentrations thresholds. Notably, it is combustion-derived particles, and not atmospheric ones that carry hazardous compounds on their surface and are especially toxic (Mills et al., 2009). There are two main ways that particles trigger a health impact: they can cause a systemic inflammation or can translocate directly into the circulation. Regardless of the pathway, the inhaled particles trigger a range of biological responses, such as elevated heart rate, blood pressure, and heart variability, which cause the onset of cardiovascular symptoms.[2]

Cardiovascular disease generates the highest costs in the German health care system and is the leading cause of deaths worldwide.[3] The studies on the link between PM_{10} and cardiovascular health consistently show the strong association between PM_{10} and heart disease, and in recent studies, particularly between cerebrovascular disease (Cesaroni et al., 2012; Lee et al., 2014; Hong et al., 2002; Tsai et al., 2003; Wellenius et al., 2005; Lisabeth et al., 2008; Stafoggia et al., 2014). Cerebrovascular disease is especially interesting, as it is the second leading cause of mortality and the leading cause of morbidity (Leiva et al., 2013). Studies also show that the burden of cardiovascular risk from pollution exposure mainly falls on the elderly and individuals with pre-existing chronic medical conditions (Hong et al., 2002; Brook et al., 2010; Wellenius et al., 2005).

The adverse association between pollution and cardiovascular health can be observed both in the short term and in the long term. The evidence for the short-term link comes from episode and time series studies, showing that variations in air pollution are paralleled by changes in cardiovascular admissions and deaths, which are observed between a few hours and two days (Pope, 2000; Wellenius et al., 2005; Fiordelisi et al., 2017). Long-term exposure studies usually evaluate health outcomes across geographical area with different levels of average pollution over longer time periods. These studies evaluate the effects of persistent exposure as well as the cumulative effects of repeated variations in average pollution levels and find robust association between exposure and disease incidence (Pope, 2000; Stafoggia et al., 2014).

This paper contributes to the existing literature in several ways. First, despite the rich evidence available on the direct link between pollution and health, less is known about the general health effect of policy instruments targeting air pollution. This paper builds on this strand of literature by providing evidence on the effectiveness of low emission zones (LEZs) as a policy instrument. A concurrent working paper by Pestel and Wozny (2019) also analyzes the effects of LEZs in Germany. The authors find

[1] Vehicle exhaust is a major contributor of carbon monoxide (CO) emissions as well. However, there are fewer CO monitors and CO concentrations are significantly below the limit values across the country.

[2] For a detailed discussion see Mills et al. (2009).

[3] According to the German Federal Statistical Office, 13.7% (46 billion euros) of health care costs in 2017 were generated by cardiovascular disease (Statistisches Bundesamt, 2017).

that LEZs reduce the number of circulatory and respiratory diagnoses among hospitals whose catchment areas have proportionally larger LEZ coverage. The present paper and the working paper by Pestel and Wozny (2019) differ in their empirical approaches, which is partly driven by the differences in the data used. Pestel and Wozny (2019) use data derived from hospital quality reports. Based on hospital location, the authors overlay its catchment area with LEZ coverage to compute the share of hospital catchment areas treated by the policy, since the quality reports do not provide information on patients' place of residence. Combining outpatient and hospital data allows me to show that it may be challenging to estimate precise treatment effects based on hospital data. I use individual-level hospital admission data, which contains information on patients' place of residence at the city level. I find that while effects point toward decreases in the number of hospital admissions, the coefficients are not precisely estimated. Supplementing the analysis with outpatient data, which allows me to locate patients' place of residence at the postal code level, I show a significant drop in the number of patients with cardiovascular disease. Encouragingly, both papers find that LEZs reduce the number of cardiovascular disease. My findings on respiratory disease, however, suggest that the number of patients with this diagnosis has not changed significantly.

A second important contribution made by this paper is linked to the fact that most of the existing literature focuses on infant and children's respiratory health, while only a small number of papers also examine the effect on the elderly.[4] In an earlier evaluation of LEZs, Gehrsitz (2017) finds no effect on infant health. Two further studies use policy-induced variation to look at infant health by exploiting electronic toll collection in New Jersey and Pennsylvania (Currie and Walker, 2011) and childhood asthma by exploiting congestion pricing in Stockholm (Simeonova et al., 2018). This paper augments the literature by providing evidence for cardiovascular disease for the entire population and for the elderly. The elderly are of particular interest here because of their susceptibility to and the prevalence of circulatory system diseases.

Third, most studies evaluating the effect of air pollution on health use temporal variations in pollution levels over a short period of time (typically a day or a week), which allows them to estimate the impact of immediate exposure, but provides little information on long term exposure effects.[5] This paper adds to the small number of existing studies analyzing the effects of long-term marginal reductions in pollution in a setting with relatively low pollution levels (Gehrsitz, 2017; Simeonova et al., 2018; Alexander and Schwandt, 2019). The nature of the present natural

[4] For papers on infant and child health, see Chay and Greenstone (2003), Neidell (2004, 2009), Currie and Neidell (2005), Currie et al. (2009), Lleras-Muney (2010), Beatty and Shimshack (2011), Coneus and Spiess (2012), Janke (2014), Sanders and Stoecker (2015), Simeonova et al. (2018) and Alexander and Schwandt (2019), and for adult health, see Beatty and Shimshack (2011), Schlenker and Walker (2016), Deryugina et al. (2016), Isen et al. (2017), Gong et al. (2019) and Anderson (2020).

[5] See, for example, Neidell (2009), Schlenker and Walker (2016) and Deryugina et al. (2016).

Table 1
Limit values for PM_{10} and NO_2 as defined by Council Directive 1999/30/EC.

	Thresholds	Deadline
PM_{10}		
Yearly average limit	40 $\mu g/m^3$	1 January 2005
Daily average limit	50 $\mu g/m^3$	
Allowed number of transgression days per year	35	
NO_2		
Yearly average limit	40 $\mu g/m^3$	1 January 2010
Daily average limit	200 $\mu g/m^3$	
Allowed number of transgression days per year	18	

Notes: Source: Council Directive 1999/30/EC Annexes II, III.

experiment and the availability of the rich and novel data over a long period enables me to evaluate health effects up to eight years after the policy was introduced. Furthermore, the national annual average level of PM_{10} in Germany was around 23 $\mu g/m^3$ in 2007 before the rollout of LEZs. Typical background concentrations of PM_{10} range between 20 and 50 $\mu g/m^3$ in developed countries and increase to between 100 and 250 $\mu g/m^3$ in developing countries (Mills et al., 2009).

The remainder of the paper is organized as follows: the next section provides the background on LEZs in Germany and Section 3 describes the data. Section 4 discusses the findings of this study for air pollution and car fleet adaptation. Section 5 presents the main results and Section 6 discusses the identification threats and the robustness checks. Finally, Section 7 concludes.

2. Low emission zones

The EU Clean Air Directives are among the strictest air quality standards worldwide. The first attempt to regulate the air quality in EU member states has been the directive 96/62/EC of the Council of the European Union, establishing the legal framework for ambient air quality regulation in all member states "to avoid, prevent or reduce harmful effects on human health and the environment as a whole". The directive defines how air quality should be assessed, specifies the pollutants for which alert thresholds and limit values should be introduced, and lists the measures member states should take to improve air quality. The daughter Directive 1999/30/EC establishes numerical limit values and alert thresholds for criteria pollutants. It divides the PM_{10} regulations into two phases: 2005–2009 and 2010 onwards with the aim to tighten PM_{10} regulations in 2010. Effective January 1, 2005, the daily average of PM_{10} concentrations must not exceed 50 $\mu g/m^3$, and the yearly average should not exceed 40 $\mu g/m^3$. For the daily threshold 35 transgressions days are permitted. For NO_2, the daily average concentrations are limited to 200 $\mu g/m^3$, with 18 transgression days and the yearly average should not exceed 40 $\mu g/m^3$. The stricter regulations have not been phased-in, as most EU members struggled to meet the 2005 regulations.[6] Table 1 summarizes these regulations.

[6] Instead EU introduced new regulations for $PM_{2.5}$ in 2008 (directive 2008/50/EC).

Between 2005 and 2007, 79 cities in Germany violated the daily threshold for PM_{10} and among them 12 violated the yearly threshold as well, according to estimations in Wolff and Perry (2010). The EU can impose significant financial penalties and even start infringement proceedings in case of non-compliance. Hence, the German government mandates that cities where even one pollution monitoring station violates the thresholds must develop a clean air action plan. Clean Air Action plans include four main elements: expanding public transportation, utilizing ring roads, improving traffic flow and, most importantly, implementing LEZs (Wolff, 2014).[7] LEZs are designated areas that ban cars from accessing the zone based on their emission class.

LEZs are phased-in with increasingly stricter restrictions. Vehicle restrictions are motivated by EU-wide tailpipe emission standards that classify passenger vehicles in categories Euro 1 to Euro 6. Cars in the highest emission category (Euro 1) receive no windscreen badge. All other cars received color-coded badges. Phase 1 bans access only to vehicles with no badges. Subsequently, Phase 2 additionally bans vehicles with red badges (Euro 2), and Phase 3 further restricts access to vehicles with yellow badges (Euro 3). Cars in categories Euro 4 to Euro 6 receive green badges and are allowed in all stages.[8] The fine for violation is 40 Euro and a driving penalty point.[9] The policy mainly affects cars with diesel engines. Cars with petrol engine receive either no badge or a green badge. However, LEZs are different from diesel driving bans, which are heavily discussed since 2018. Diesel bans restrict access to all diesel cars, with the exception of diesel Euro 6 category.[10]

Fig. 1 plots the timing of these three phases. There is a vast temporal and spatial variation in the introduction of zones. The first zones were enacted in 2008. Among early introducers are Berlin, Hannover and Cologne, as well as cities in the Ruhr area, which in January 2012 united into a common LEZ for the Ruhr area, covering 869 km^2 and including 2.5 million residents. Not all LEZ-cities are non-attainment cities and vice versa: a number of non-attainment cities refrained from implementing LEZs.[11] Fig. 2 illustrates the sample composition as well as spatial and temporal variation that I exploit in the empirical design. White perimeters represent the large cities which by 2017 have had no LEZ. The color-coded perimeters represent the large cities that introduced an LEZ in the observation period. Each color represents a group that enacted the zone in the given year. Towns and rural areas, shaded in gray, are excluded from the sample.

Table A.3 in Appendix A presents the detailed list of the LEZ cities, with the enactment date, the areal coverage, and the attainment status of treated cities. Since each city is responsible for designing, enacting, and enforcing own LEZs, there is large variation in the proportional size of LEZs: the coverage ranges from as little as 1% of the city area to just below 100% covering the entire city. The perimeters are not randomly drawn—they mainly cover (potential) non-attainment areas of cities. Additionally, factors such as composition of the local fleet, urban layout, and social justice matter for designing the zones. A descriptive analysis suggests that in treated cities LEZs on average have slightly lower income compared to the rest of the city, have less cars per household and fewer residents over the age of 65. The most prominent difference between LEZs and the rest of the city is the population density, which is markedly higher inside the zones.[12]

Previous evaluations of LEZs in Germany all suggest that the zones reduce PM_{10} concentrations in treated cities. The first evidence comes from Wolff (2014), who analyses the short-term effects of LEZs until October 2008. He finds around a 7–9% drop in daily PM_{10} levels. Further evaluations support these initial findings, despite differences in time span and in composition of treatment and control units (Morfeld et al., 2014; Malina and Scheffler, 2015; Gehrsitz, 2017; Jiang et al., 2017). In the most recent evaluation, Gehrsitz (2017) finds around a 2.5% reduction in daily PM_{10} levels on average, and around a 1.5% decrease in daily NO_2 levels. Gehrsitz (2017) additionally analyses potential effects on infant health, measured by birth weight, the incidence of low birth weight, and still birth. He finds a null effect.

It is worth noting that the diesel scandal happened in the middle of the observation period of this study. It is known now that Volkswagen and a few other car manufacturers have been manipulating the actual emissions of cars in laboratory settings through a software. The cars with clean diesel engine were introduced in 2008 and by 2017 there were 5.3 million such cars in Germany. Officially, these cars were mostly in Euro 5 and Euro 6 categories. The Federal Motor Transport Authority of Germany estimates that, on average, the cheating cars were emitting 59% more on the streets than their laboratory tests claimed. Alexander and Schwandt (2019) show the significant impact of the emission cheating on air pollution and health in the US. Due to this, it is likely that the pollution effects of LEZs shown in this paper are a lower bound.

3. Data

To study the effect of LEZs on air pollution, I combine data from multiple sources. I obtain daily PM_{10} and NO_2 measurements from the German Environment Agency (Umweltbundesamt) for the period 2004–2016. The sample consists of 264 PM_{10} monitoring stations and 261 NO_2

[7] Wolff (2014) shows that the other elements of APs in general do not have a significant effect on PM_{10}.

[8] Table A.1 in Appendix A presents details on tailpipe emission regulations. Table A.2 presents some typical examples of cars in each Euro-class.

[9] The drivers lose their license after accumulating 18 points. However the fine for LEZ violation was replaced with 80 Euro and no penalty points in May 2014.

[10] Diesel bans are stricter than LEZs, but are rather unpopular in Germany. The biggest issue is with enforcing compliance. The problem is that currently all Euro 4, 5 and Euro 6 categories receive a green windshield badge, meaning that these cannot be differentiated either by the badge or the license plate. The only proof whether a vehicle is diesel is in the registration document. Hence, enforcement would require stopping vehicles for checks.

[11] Attainment and non-attainment in this context mean cities that comply with air pollution regulations and cities that do not.

[12] For details, see Table A.4 in Appendix A.

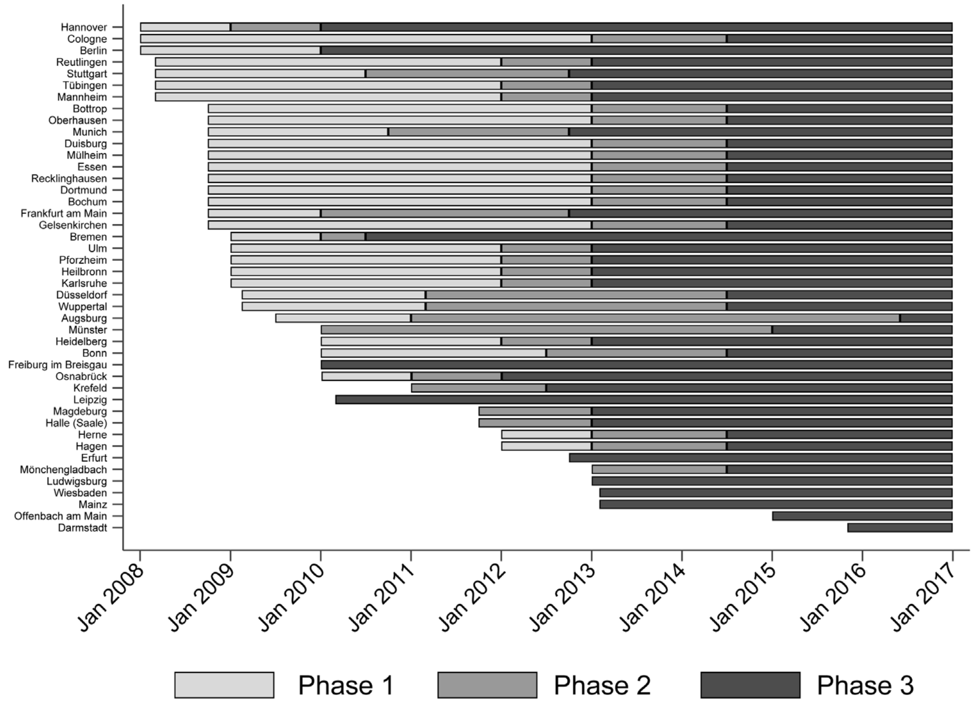

Fig. 1. LEZ introduction and phases. *Notes:* The zones are introduced in three phases. Phase 1 restricts access to cars that receive no windscreen sticker. Phase 2 restricts access additionally to cars with red windscreen stickers, and Phase 3 – additionally to yellow sticker cars. The color-coded stickers are given based on emission classes. Some of the cities, which introduced LEZs in 2011 or later, enacted zones directly as Phase 2 or Phase 3.

monitoring stations in 69 cities. Using the precise geographic coordinates of each station, I locate whether it is inside or outside an LEZ.[13] To account for weather-pollution interaction, I additionally collect weather data from the German Weather Service (Deutscher Wetterdienst, DWD). The data contain information on daily temperature, wind speed, precipitation, cloud cover, vapor, air pressure, and relative humidity. I match each air quality monitoring station to its geographically closest weather station and aggregate the daily measurements at the monitoring station-month level.

To study the health impact of LEZs, I draw on outpatient health data from the Central Research Institute of Ambulatory Health Care (Zi). These data comprise nationwide ambulatory care claims for patients of all statutory health insurance funds in Germany. The observations are annual and cover the period 2009–2017. This is a novel data source, which contains information on patients' postal code of residence. Because of data protection laws, it is not possible to obtain data on the postal code level directly. Instead, I obtain the number of patients with the diagnosis of interest, aggregated separately for control cities and by the areas inside and outside a low emission zone for treated cities. The diagnoses are coded according to the 10th revision of International Classification of Disease (ICD-10). Fig. 3 illustrates the observation level with the example of Berlin. The gray shaded area covers the LEZ in Berlin. Hence, for Berlin, I obtain yearly aggregated data for the gray and white shaded areas separately. The advantage of this breakdown is that the treatment status can be defined very precisely. For cities that have no LEZ, I obtain the aggregated number of patients for the city as a whole.

My main outcome is the number of patients with cardiovascular disease (I00–I99). The data also allow to separately analyze heart disease (I20–I49) and cerebrovascular disease, including stroke (I60–I66), and to zoom into certain age groups. Of central interest are the elderly above the age of 65, however I also show results for the age groups 15–29 and 30–64. The data for younger patients is less reliable: the data confidentiality requires censoring all cells with less than 30 observations, which for patients under the age of 15 results in a large number of missing values.

I complement the main analysis with additional data from the hospital diagnosis statistics of the Federal Statistical Office. The inpatient administrative register comprises 70% random sample of all hospital admissions in 2004–2014, including emergency room admissions without overnight stay. The data provide information on date of admission and discharge, primary diagnosis, city of residence, as well as age and gender. However, it has two major caveats: first, identifying whether the patient lives inside or outside an LEZ is not possible. Hence, I define the treatment status by city of residence. Second, hospital admission is a severe outcome, particularly so in Germany, where hospitals are obliged to justify that an outpatient treatment is insufficient.

Panel A of Table 2 shows the means of PM_{10}, NO_2 and inpatient health outcomes in 2007 and reports the two

[13] The geographic coordinates are provided by the German Environment Agency. To classify the stations, I rely on open source polygons of Low Emission Zones from OpenStreetMap.

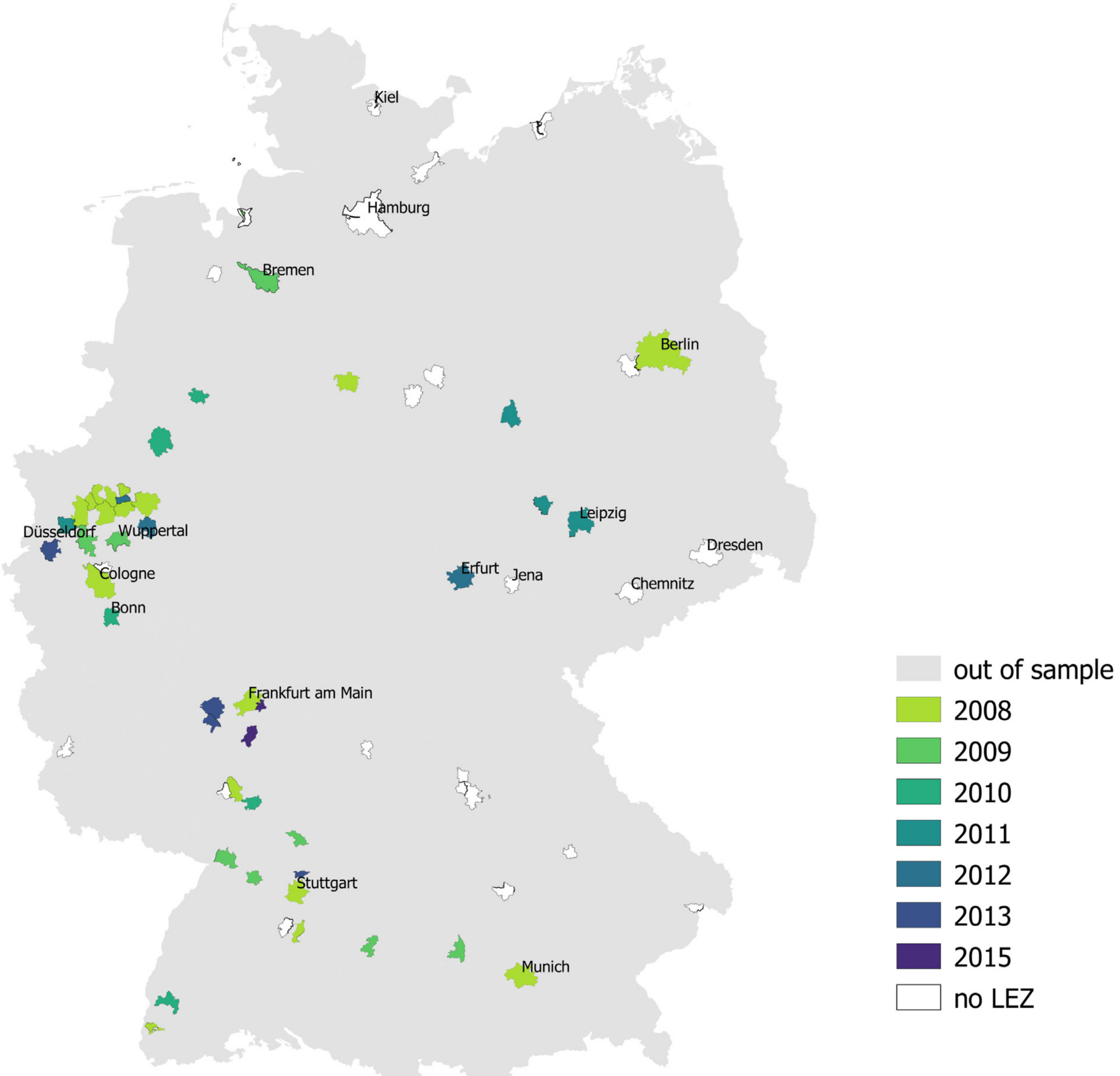

Fig. 2. The variation in enactment of LEZ in major German cities. *Notes:* The map displays the untreated and treated cities, classified by the year of LEZ introduction. Some city names are displayed for ease of geographic orientation. Gray shaded areas are not part of estimation sample.

sided *p*-value of the null hypothesis that the levels of these variables were the same in treated and untreated cities. Since LEZs are mainly measures employed in urban areas, I restrict the sample to cities with more than 100,000 residents to increase comparability between treated and untreated units. In this sample, the treated units are the cities that have introduced an LEZ during the observation period, and the control units are cities that have not introduced an LEZ by the end of the observation period.[14] I choose 2007, as it is the last year before the first rollout of LEZs. For proxying the baseline differences in health outcomes, I use inpatient data, as the outpatient data is available only starting in 2009.

The annual average levels of PM_{10} and NO_2 in 2007 are, as expected, different between treated and untreated cities. LEZ cities have on average higher PM_{10} and NO_2 concentrations in 2007, and these differences are statistically significant. Note however, that even in non-attainment cities the annual concentrations of both pollutants are relatively low. Hence, results should be interpreted in the context of reductions against already low reference levels. In contrast to pollution, the prevalence of cardiovascular disease, measured as the number of hospital admissions per 10,000 inhabitants, appears to be statistically indistinguishable between treated and untreated cities.

[14] Note that the introduction of the zones continues and by 2019, many of the control units have enacted their LEZs as well.

Fig. 3. Berlin LEZ. *Notes:* The map illustrates the Berlin LEZ (gray shaded) and the area outside (white shaded).

Table 2
Means and pre-trends of air quality and health outcomes.

	PM_{10} (1)	NO_2 (2)	Cardiovascular (3)
Average levels in 2007			
Untreated	23.4	34.08	9.51
Treated	27.02	42.23	9.31
p-value	0.00	0.00	0.87
p-value on difference in pre-trends			
2005–2004	0.91	0.57	0.89
2006–2005	0.01	0.07	0.01
2007–2006	0.27	0.13	0.22

Notes: Variable names in headings. PM_{10} and NO_2 are measured in $\mu g/m^3$. Cardiovascular disease is measured as the number of admissions per 10,000 inhabitants. The lower panel presents the p-value on the difference in changes between treated and untreated cities. *Data*: German Environment Agency and RDC of the Federal Statistical Office and Statistical Offices of the Federal States, Hospital statistics, 2004–2014.

It is straightforward to control for baseline differences, as long as the main identifying assumption holds. That is, that the trends between LEZ and non-LEZ cities do not differ systematically for reasons other than the implementation of the zones. To provide a suggestive evidence towards this assumption, Table 2 reports the *p*-values from a test of the null hypothesis that the year-on-year changes in all outcomes are different from zero. The results show that in the majority of cases the yearly changes in outcomes are not statistically different between treated and untreated cities before 2008. Event studies in Sections 4 and 5 also show that pre-trends do not vary over time.

4. The effect of LEZs on air pollution and car fleet composition

I first investigate how LEZs affect air pollution and car fleet composition, and then turn to health outcomes. Again, I restrict the sample to cities with more than 100,000 residents. To evaluate the effect of LEZs on pollution, I estimate the following equation:

$$y_{ict} = \alpha + \beta LEZ_{ict} + M_t + S_i + W_{ict} + \epsilon_{ict} \tag{1}$$

where y_{ict} is the monthly average concentration of PM_{10} or NO_2 at station i at time t in city c. LEZ_{ict} indicates whether a station i is located inside an LEZ at time t in city c. M_t and S_i are year-month and monitoring station fixed effects. Including station fixed effects accounts for time-invariant differences in the level of pollution between the stations (and, hence, also between the cities as the stations are nested in cities) and ensures that identification comes from within-station variation over time. Year-by-month fixed effects net out time shocks that commonly influence pollution in the cities. Furthermore, the vector W_{ict} includes a set of controls for weather, in particular, average temperature, its quadratic, maximum and minimum temperature, average humidity and its quadratic, an interaction term between average temperature and humidity, average air pressure, average precipitation and its quadratic, average wind speed, a dummy variable indicating rainfall, and an interaction term between average wind speed and average temperature. ϵ_{ict} is the error term. I cluster standard errors

Table 3
The effect of LEZ on monthly PM_{10} and NO_2 concentrations.

	Monthly PM_{10}			Monthly NO_2		
	(1)	(2)	(3)	(4)	(5)	(6)
In LEZ	−0.872**	−0.622*	−0.869**	−0.818*	−0.748	−0.557
	(0.357)	(0.327)	(0.329)	(0.466)	(0.492)	(0.565)
Pre-LEZ mean	27.88	27.88	27.88	45.05	45.05	45.05
Change in percent	3.12	2.23	3.12	1.82	1.66	1.24
Observations	26,523	25,874	17,974	27,637	26,945	19,224
Station fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Weather controls	No	Yes	Yes	No	Yes	Yes
Station restrictions	No	No	Yes	No	No	Yes

Notes: The outcome is the monthly concentration of either PM_{10} or NO_2. Pre-LEZ mean refers to average concentrations of the respective pollutant at stations inside an LEZ before the enactment of the zone. Results from Eq. (1). Columns (1)–(2) and (4)–(5) include all stations. Columns (3) and (6) exclude stations in treated cities that are not inside an LEZ. Robust standard errors, clustered at the city level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
Data: German Environment Agency and German Weather Service.

at the city level to allow for serial correlation within cities over time.

I estimate a variant of Eq. (1) excluding all pollution monitoring stations located in a treated city, but outside a low emission zone. These stations might be subject to negative spillover if drivers take longer tours to avoid LEZs, or positive pollution spillovers if the introduction of zones leads to a change in fleet composition or a drop in car usage in general.

Table 3 reports the results for PM_{10} in columns (1)–(3) and for NO_2 in columns (4)–(6). The results suggest that LEZs reduce monthly PM_{10} concentrations by 0.6–0.9 $\mu g/m^3$. The estimated coefficient is 0.9 $\mu g/m^3$ in column (3). This translates into a 3% decline in monthly PM_{10} relative to the average concentration levels in pre-LEZ period at treated stations. The findings for PM_{10}, being comparable in column (2) and (3), suggest negligible spillovers in either direction. The findings for NO_2 suggest a small reduction in its monthly concentration. After controlling for weather covariates, however, the effect becomes statistically insignificant.[15]

I focus on PM_{10} and not $PM_{2.5}$ as the number of $PM_{2.5}$ monitoring stations has been very small throughout the observation period, particularly for the period 2004–2010. Table A.5 in Appendix A presents the results for $PM_{2.5}$. The findings show that $PM_{2.5}$ decreased by around 0.47 $\mu g/m^3$.

To capture the dynamic effect of LEZs on pollution and gain more insights on the difference between treated and untreated cities in the pre-treatment period, I estimate an event-study model as follows:

$$y_{ict} = \alpha + \sum_{k=-5, k \neq -1}^{5} \beta_k LEZ_{ik} + M_t + S_i + W_{ict} + \epsilon_{ict} \qquad (2)$$

where the dummy variables LEZ_{ik} indicate yearly lags and leads of up to five years before and after the introduction of LEZs. The reference category is the period −1, hence the effects are relative to the year immediately before the enactment. The rest of the controls are as specified in Eq. (1). Fig. 4 presents the coefficients from the event study. The plots suggest that the common trend assumption is likely to hold, as all pre-LEZ coefficients are close to zero and statistically insignificant. The pre-LEZ coefficients are also jointly statistically insignificant. The respective p-values of an F-test for the joint hypothesis that all pre-LEZ coefficients are zero, are 0.74 and 0.63 for PM_{10} and NO_2. The event plots also suggest that the reductions in pollution levels are stronger from the third year onward, compared to the year just before the enactment of the zones.

These results are broadly in line with previous findings (Wolff, 2014; Gehrsitz, 2017). The lack of impact on NO_2 may reflect the slow improvements in nitrogen oxide emissions in real world conditions by diesel engines. Carslaw and Rhys-Tyler (2013) show that the reduction in NO_x emissions from all types of diesel vehicles over the past 15–20 years has been very small.

4.1. Change in car fleet composition as a potential channel

There might be multiple channels through which LEZs affect the air quality. In the short term, LEZs are likely to reduce the absolute number of cars on the roads. Car owners with vehicles that do not comply with the regulations might switch to different means of transportation, such as public transport or bicycles. In the medium and long term, however, the absolute number of cars might increase again. Thereby, the composition of the fleet with respect to emission standards will likely be different, that is, the highest emitting cars will be replaced with cleaner ones.

This section studies the effect of LEZs on car composition. I draw on data from the Federal Motor Transport Authority (Kraftfahrtsbundesamt Flensburg). The data includes yearly observations on the total number of passenger vehicle registrations by city and emission class (Euro 1–Euro 6). I collect data for the period 2008–2017. The data

[15] In unreported regressions I also run specifications where the air pollution is aggregated at the city level and separately by traffic stations and by background stations, as defined by the German Environment Agency. The results confirm a significant 1.2 $\mu g/m^3$ reduction in PM_{10} at the city level across traffic stations. The reductions across background stations and in NO_2 are small in magnitude and statistically not different from zero.

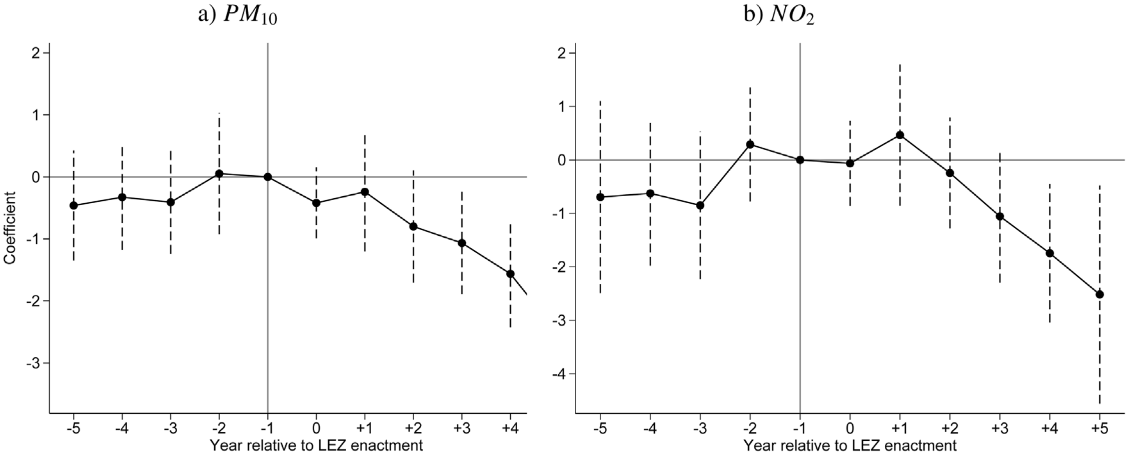

Fig. 4. Event study of annual concentrations of PM_{10} and NO_2. *Notes:* The outcome variable is the yearly average concentration of PM_{10} and NO_2. Coefficients and 95% confidence intervals from Eq. (2). The vertical line refers to $t = -1$ as the reference year. Standard errors are clustered at the city level. *Data:* German Environment Agency and German Weather Service.

is reported on January 1 of each year, hence the factual period the observations cover is 2007–2016. Note that the assigned color of the windshield stickers mainly depends on the Euro-class of the car, but can vary depending on the tax class of the car and the existence of a particle filter. Given this shortcoming of the data, it is likely that the share of no-, red-, and yellow-sticker cars is lower in practice, while the share of green-sticker cars is probably higher. Following the implementation of LEZs, the composition of the car fleet has changed significantly. As Fig. A.2 in Appendix A shows, the share of Euro 1 cars has dropped from around 20% in 2007 to below 2% in 2017. In the meantime, the share of Euro 4 and higher class cars has increased from 40% to nearly 80%.[16]

To study the effect of LEZs on the composition of vehicle registration more formally, Table 4 reports the regression results following the same difference-in-differences empirical approach as in the previous section. The outcome variable is the share of the respective class of cars as specified in column headings in the total number of registrations in a city. The results support that LEZs reduce the share of Euro 1 cars and increase the share of Euro 4 cars in treated cities. The effect of LEZs on the share of Euro 4, 5 and 6 classes combined is positive, but statistically insignificant. This is conceivable, given that Euro 5 and Euro 6 categories, despite receiving a green badge, were only introduced as classes in September 2009 and September 2014 respectively. Panel B replaces the binary LEZ indicator with three indicators for each of the phases to zoom into the effect of each phase separately. The results suggest that the largest reduction in Euro 1 cars happens in Phase 2 and 3. The share of Euro 4 cars increases most strongly in Phase 3. These results are line with Wolff (2014), who also finds that LEZs create an incentive for drivers to substitute towards lower emitting vehicles.

[16] Fig. A.3 in Appendix A illustrates this development for each city separately.

5. The effect of LEZs on cardiovascular health

This section presents the main findings of the paper. To analyze the outpatient data I estimate the following regression specification:

$$\log(y_{at}) = \alpha + \beta LEZ_{at} + transition_{at} + T_t + D_a + \epsilon_{at}, \quad (3)$$

where the outcome is the number of patients with the given diagnosis in logarithms in area a in year t. LEZ_{at} indicates whether the area a has LEZ in January of year t. While most cities enact their LEZs at the beginning of the year, some introduce theirs later. Thus, I separate those periods by including the dummy *transition* that is one for all years when the zone was active for less than 12 months. T_t and D_a are year and area fixed effects. In this equation, area refers to either an LEZ, the peripheries of treated cities outside LEZs or an untreated city. Including area fixed effects accounts for time-invariant differences and ensures that identifying variation comes from within-area variation over time. It also takes into account general differences in size of the areas. Year fixed effects net out common time shocks and trends that affect all areas similarly.

As Section 3 discusses, the main caveat of the outpatient data is missing observations before 2009, while the rollout of LEZs started already in 2008. This data restriction is particularly meaningful if the effect of LEZs on health is time varying. In a recent working paper, Goodman-Bacon (2018) shows that in difference-in-differences framework with timing variation, every unit acts as a control unit at some point. Hence, treatment effects that pick up over time in the post-treatment period will lead to a downward bias in the difference-in-differences coefficient. Therefore, I also present the results based on the sample that excludes previously treated cities from the estimation. Section 5.2 shows the event dynamics of the coefficients and implements the decomposition proposed in Goodman-Bacon (2018). Section 5.3 further juxtaposes the pollution and health effects in this restricted sample and provides more insights into the timing of the effects.

Table 4
The effect of LEZ on the share of cars by emission class.

	Euro 1 (1)	Euro 2 (2)	Euro 3 (3)	Euro 4 (4)	Euro 4, 5 and 6 (5)
Panel A: Single LEZ indicator					
LEZ	−0.354**	−0.329	0.119	1.891*	0.564
	(0.152)	(0.340)	(0.181)	(1.119)	(0.376)
Panel B: Indicator for each phase					
Phase 1	−0.132	−0.154	−0.011	1.172	0.296
	(0.116)	(0.274)	(0.166)	(0.874)	(0.269)
Phase 2	−0.408**	−0.395	0.208	1.921	0.595
	(0.166)	(0.361)	(0.190)	(1.342)	(0.390)
Phase 3	−0.540**	−0.480	0.221	2.560*	0.799
	(0.216)	(0.432)	(0.250)	(1.368)	(0.534)
Observations	690	690	690	690	690
Year FE	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes

Notes: The outcome is the share of cars in each emission class in the entire registration of cars annually in a given city. Panel A controls for a single binary indicator for LEZ. Panel B includes binary indicators for each of the phases. Robust standard errors, clustered at the city level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Data:* Federal Motor Vehicle Authority.

5.1. Main results

Table 5 presents the main results for cardiovascular disease. Columns with odd numbers present estimates from the full sample. Columns with even numbers present the estimates from the restricted sample. Hereby, the treatment sample includes those cities that introduce LEZs after 2009.

Panel A presents the effect of LEZ on all cardiovascular disease. In column (1) the number of patients decreases by 1.2%, however the coefficient is imprecisely estimated. After restricting the sample, column (2) shows that the number of patients with cardiovascular disease of all ages decreases by 2.2% over the post-treatment period. This reduction translates into 0.9 fewer patients yearly per 10,000 people. Columns (3) and (4) present the effect for the elderly above 65. The point estimates suggest a 2–3% reduction in the number of patients. This translates into approximately 10 fewer patients yearly per 10,000 elderly.

In Panel B and C, I further slice the cardiovascular diagnoses into two subgroups: heart disease (I20–I49) and cerebrovascular disease (I60–I66). I focus on these two subgroups, as they are the most common cardiovascular diseases, accounting for a large share of costs, morbidity and mortality (Leal et al., 2006; Restrepo and Rieger, 2016). Together heart disease and cerebrovascular disease account for around 80% of all cardiovascular disease in the sample.

The estimates suggest that the decrease in heart disease is statistically significant only in the restricted sample. Here the effects are pronounced for those over the age of 65. The effects on cerebrovascular disease are much larger in magnitude and suggest a reduction of 7–13% both for the overall population and the elderly.

To compare the effect of LEZs across age groups, Fig. 5 plots the coefficient from the restricted sample for age groups 15–29 and 30–64, juxtaposing these coefficients to the effect for population over 65. The prevalence of cardiovascular disease increases strongly with age. The fraction of people with a cardiovascular diagnosis is very low for those below the age of 30 (2% at most). The prevalence increases

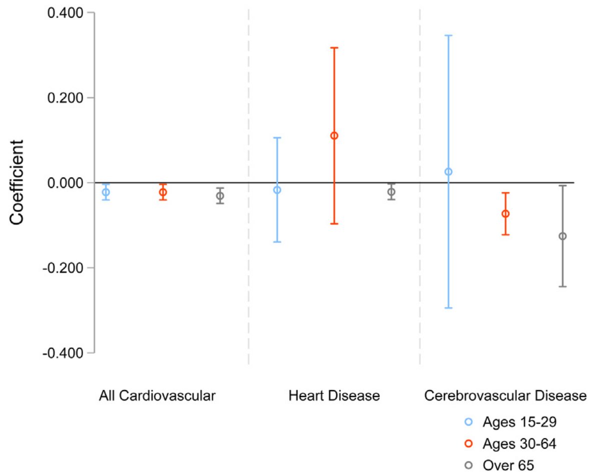

Fig. 5. The effect of LEZ on cardiovascular health, by sub-diagnoses and age groups. *Notes:* The model specification comes from Eq. (3). The figure presents results for three age groups from the subsample of cities that introduce LEZs after 2009. Coefficients and 95% confidence intervals. Robust standard errors are clustered at the city level. *Data:* Zi.

in the age group 30–64, and is rather high for those over the age of 65. The prevalence of heart disease and cerebrovascular disease is noticeably different. For example, 6.8% of 30–64 year olds have a heart disease diagnosis, but only 1.4% have a cerebrovascular disease. The pattern is similar for the elderly, albeit the respective numbers – 42.4% and 11.1% – are much larger.

The figure shows that the number of all cardiovascular diagnoses decrease for all age groups, however, the largest reductions are indeed observed for the elderly patients. The pattern is different when separating the diagnosis group into heart disease and cerebrovascular disease. The estimates suggest that heart disease improves only for the elderly, while cerebrovascular disease improves for middle-aged adults as well.[17]

[17] I cannot reject the null hypothesis that the two coefficients are equal to each other. The age-specific effects are not thoroughly discussed in

Table 5
The effect of LEZ on cardiovascular disease.

	All age groups		Elderly above 65	
	(1)	(2)	(3)	(4)
Panel A: All cardiovascular disease				
LEZ	−0.012	−0.022**	−0.021**	−0.031***
	(0.010)	(0.009)	(0.009)	(0.009)
Pre-LEZ mean	9.81	9.81	9.48	9.48
Observations	954	585	954	585
Panel B: Heart disease				
LEZ	−0.006	−0.012	−0.016	−0.021**
	(0.010)	(0.009)	(0.011)	(0.009)
Pre-LEZ mean	9.25	9.25	8.88	8.88
Observations	954	585	954	585
Panel C: Cerebrovascular disease				
LEZ	−0.072**	−0.126**	−0.071*	−0.126**
	(0.036)	(0.059)	(0.036)	(0.059)
Pre-LEZ mean	7.86	7.86	7.56	7.56
Observations	954	585	954	585
Area FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Intro after 2009	No	Yes	No	Yes

Notes: The outcome is number of patients with the given diagnosis in logarithms. The restriction "Intro after 2009" refers to keeping only those treated cities that introduced an LEZ after 2009. Pre-LEZ mean refers to the logarithm of average number of patients in the entire pre-treatment period across cities. Robust standard errors, clustered at the city level, in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Data:* Zi.

Considering prior work, the magnitude of the effects appears plausible. The German Environment Agency reports that multiple modeling studies initially suggested that LEZs can potentially reduce yearly average concentrations of PM_{10} by 2% in Phase 1 up to 10% in Phase 3 compared to pollution concentrations in 2007 (Diegmann et al., 2007). I estimate a 1.1 $\mu g/m^3$, which is a 4% decrease in PM_{10} concentrations annually compared to 2007. This number, although closer to the lower bound, is within the interval of initial expectations.

Benchmarking the effect of LEZs on cardiovascular disease is not straightforward as, to the best of my knowledge, there are no quasi-experimental studies that link PM_{10} directly to cardiovascular disease. Hence, I rely on a prospective cohort study and a meta-analysis in 11 European cohorts by Cesaroni et al. (2012). The authors show a 10 $\mu g/m^3$ increase in annual mean PM_{10} is associated with 12% increase in cardiovascular events. The association persists also at low levels of exposure. A linear extrapolation implies that for a 1.1 $\mu g/m^3$ decrease in PM_{10} I should expect approximately a 1.3% drop in cardiovascular events. Estimates in Table 5 suggest a 2% reduction in cardiovascular events in general population and a 3% decrease for the elderly. My estimates are thus slightly larger, which might indicate that prior work underestimated the effect of PM_{10} on cardiovascular outcomes. Previous studies also suggest that the risk due to 1 $\mu g/m^3$ increase in PM_{10} is roughly comparable to the excess cardiovascular risk of smoking one cigarette a day, which is around 4% (Law et al., 1997).

the literature. Many epidemiological studies focus on the older individuals (most often over age 65) and exclude younger populations from the beginning. Studies that stratify the analysis by age groups find contradicting results. Some do in fact conclude that age has a strong effect on cerebrovascular health (Linn et al., 2000; Hong et al., 2002; Zhang et al., 2011; Maheswaran et al., 2012), while others find effects independent of age (Lee et al., 2018; Chau and Wang, 2020).

A back-of-the-envelope cost-benefit analysis suggests that the benefits generated by the lower number of patients with cardiovascular disease surpass the costs that accrued due to induced vehicle upgrade by around 2 billion Euro. Appendix C provides the details of the calculation.

5.2. Event study and Goodman-Bacon decomposition

The estimated coefficients in Table 5 report the effect of LEZ averaged over the entire study period. However, the LEZ-induced pollution reductions may have a lagged effect on health that may also change over time. Hence, I estimate an event-study model as follows:

$$\log(y_{at}) = \alpha + \sum_{k=-3, k\neq -1}^{4} \beta_k LEZ_{ak} + T_t + D_a + \epsilon_{at} \tag{4}$$

Fig. 6 presents the event study graphs for all cardiovascular diagnoses for all ages and the elderly over 65 separately. Each panel presents the coefficients and their 95% confidence intervals from the entire sample (red line) and after restricting the sample to cities treated after 2009 (blue line). The left panel plots the results for the entire population, and the right panel – for the elderly. Both graphs show that the trends in cardiovascular disease between treated and untreated cities display no clear trend before the implementation of LEZs. The pre-treatment coefficients are also jointly statistically insignificant. The respective p-values of the F-test for the hypothesis that all pre-treatment coefficients are jointly zero are 0.24 (entire sample) and 0.15 (introduction after 2009) in the left panel, and 0.14 for both samples in the right panel. Upon LEZs' enactment, the number of patients falls in treated cities at a faster rate than in untreated cities. Both graphs also show that the treatment effects are not time-constant:

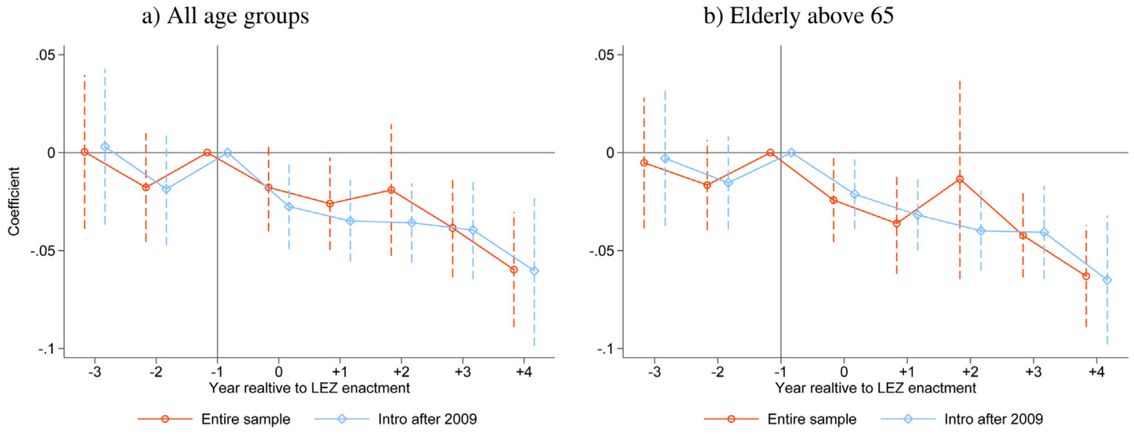

Fig. 6. Event study of cardiovascular disease. *Notes:* The outcome variable is the logarithm of the number of patients with cardiovascular disease. The red line reports the coefficients from a regression on the entire sample. The blue line restricts the sample to cities that have introduced an LEZ after 2009. Coefficients and 95% confidence intervals from Eq. (4). The vertical line refers to $t = -1$ as the reference year. Standard errors are clustered at the city level. *Data:* Zi. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

their absolute magnitude tends to become larger the longer LEZs are in place.

Time-varying treatment effects might bias the difference-in-differences estimate away from the true effect, as discussed in Goodman-Bacon (2018). The author shows that in difference-in-differences designs with timing variation, the difference-in-differences regression coefficient is a weighted average of all possible difference-in-differences coefficients of two-group two-period (2x2) comparisons, where the weights depend on the sample share and the treatment variance in each pair. It is possible to decompose and visualize each of these 2x2 estimates against their weight. The decomposition illustrates how average estimates vary across types of comparisons and which comparisons matter most.

To illustrate the proposed decomposition, Fig. 7 plots the difference-in-differences estimates for cardiovascular health in the present setting. The graphs refer to all cardiovascular diagnoses from the entire sample for the overall population and the elderly. The vertical axis plots the 2x2 estimate for each pair and the horizontal axis plots the weight each of these pairs receive. The horizontal line shows the weighted average of all difference-in-differences estimates. The figure highlights the influential role of the pair “treatment versus never treated”. In both panels, 68% of the variation comes from this comparison. This is not coincidental, as the variation share reflects the sample shares and the treatment variance, which are identical for both outcomes. The pure timing group comparisons get very small weights (2.5% for “earlier group treatment versus later group control”, and 6.4% for “later group treatment versus earlier group control”). The figure also illustrates that the regression difference-in-differences coefficients might be smaller in magnitude due to time-varying effects. As the hollow circles show, the treatment versus already treated comparison mostly generates positive difference-in-differences coefficients with non-negligible weight (23.6%). It is possible to take out the bias from time-varying effects by subtracting the weighted average of all 2x2 difference-in-differences comparisons where the controls are the already treated units. This provides insight into why the difference-in-differences coefficients in Table 5 become larger in magnitude after restricting the treatment sample to cities that introduce LEZs after 2009. This empirical exercise supports the main findings and motivates the restricted specification in column (2) of Table 5 as the preferred specification.

5.3. The timing of pollution and health effects

This section examines the timing of changes in pollution and cardiovascular disease by unpacking the event dynamics and the separate effects of three LEZ phases.

The analysis of health data shows that the effects are strongest when the sample is restricted to treated cities that have introduced LEZs after 2009, as Fig. 6 shows. Here I use the same restricted sample to demonstrate that in this subsample as well LEZs reduce air pollution. Furthermore, to gain insights into the timing of pollution and health effects, Fig. 8 draws the event dynamics in this subsample for PM_{10}, NO_2 and cardiovascular disease.

There are two main takeaways from the event plots in Fig. 8. First, LEZs reduce PM_{10} also in the subsample of post-2009 LEZ cities, whereas the reductions in NO_2 remain imprecisely estimated. Second, the reductions in the number of patients with cardiovascular diagnoses and in PM_{10} are congruent. The graph suggests that both PM_{10} and cardiovascular disease respond already in the implementation year of LEZs. This is not surprising. A large number of studies show that the impact of air pollution on health can be observed both in the short term and in the long term. Studies evaluating short-term variations in air pollution show that cardiovascular admissions and deaths respond already within a few hours up to two days (Pope, 2000; Wellenius et al., 2005; Fiordelisi et al., 2017).

The implementation of LEZs included several phases, as Section 2 discusses. These phases may affect the concentration of pollutants, and hence health outcomes differently. To separate the effects of the different phases, I replace the LEZ indicator with three indicators for each of the phases in Eqs. (1) and (3), such that each indicator captures the effect of that phase individually. Fig. 9 illustrates the coefficients

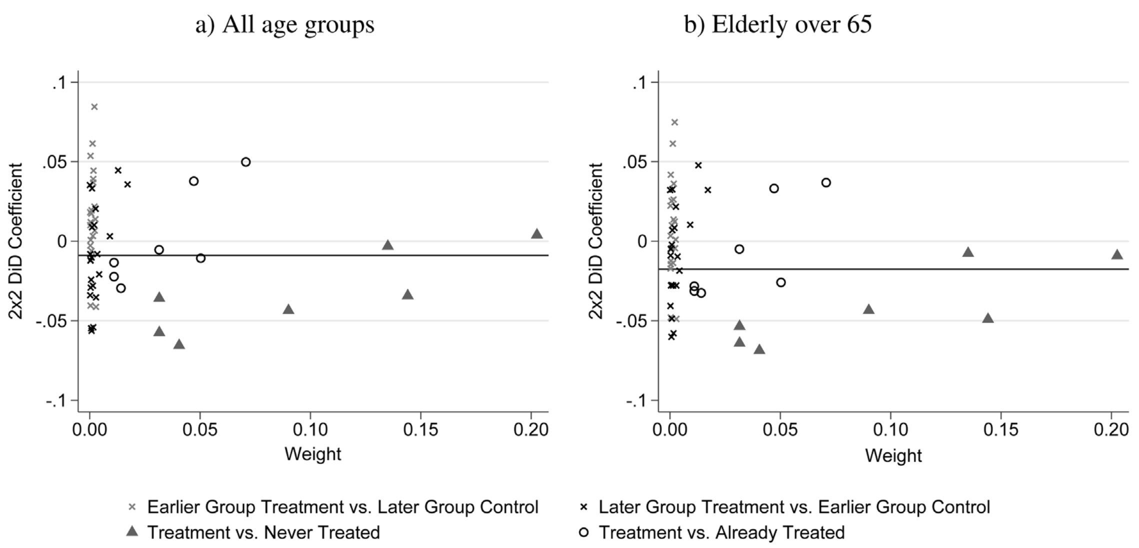

Fig. 7. Difference-in-differences decomposition for cardiovascular disease. *Notes:* The figures plot each 2x2 difference-in-differences components from the Goodman-Bacon (2018) decomposition theorem. The horizontal line signifies the average DiD estimate, and equals to the sum of *y*-axis values weighted by *x*-axis values. *Data:* Zi.

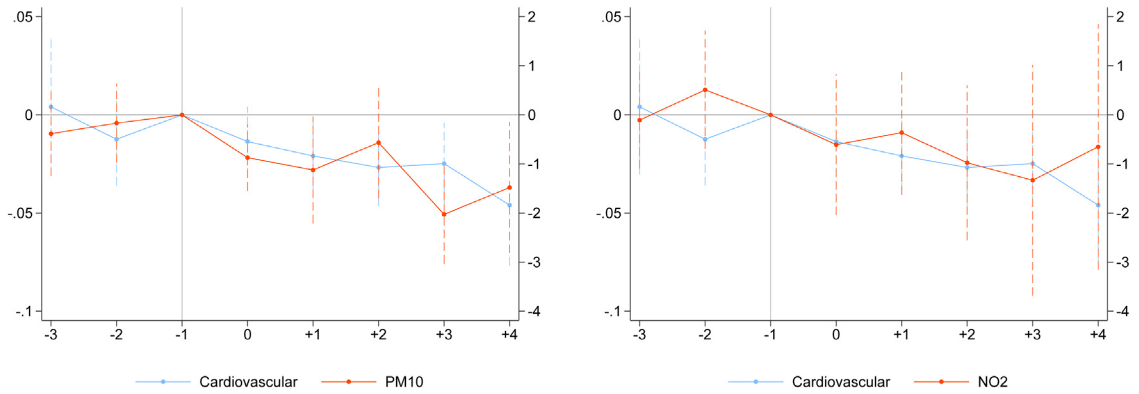

Fig. 8. Event study of cardiovascular disease and annual concentrations of PM_{10} and NO_2. *Notes:* The figure presents coefficients from a subsample of treated cities that introduce LEZs after 2009. The vertical line refers to $t = -1$ as the reference year. Coefficients and 95% confidence intervals come from Eqs. (2) and (4). Standard errors are clustered at the city level. *Data:* Zi, German Environmental Agency and German Weather Service.

and confidence intervals of these three phases. To juxtapose pollution and health effects, each graph combines the cardiovascular outcome under investigation with PM_{10} and NO_2. As in Table 5, I look at heart disease and cerebrovascular disease, as well as the elderly above 65 separately.

The results suggest that Phase 2 has the strongest impact on both pollutants. However, while the decreases of PM_{10} in Phase 2 and Phase 3 are comparable, the decrease in NO_2 is large in Phase 2, but bounces back in Phase 3. This result squares with two potential explanations. It is possible that Phase 2 achieves the largest environmental impact, as the number of yellow sticker cars on the roads is large. It is also possible that in this phase in most cities there were less cars on the roads, if the vehicle fleet has not managed to fully adapt yet. The pattern is different for health outcomes: the decrease in the number of patients becomes slightly larger in each phase. This development is suggestive evidence for a buildup in health effects.

5.4. Further health outcomes

A careful reading of epidemiological and economic literature suggests that respiratory disease, diabetes and central nervous system (CNS) disease are also frequently linked to air pollution. Economic literature has extensively investigated respiratory disease in particular. Most papers show a negative relationship between air pollution and respiratory health, especially for children (Lleras-Muney, 2010; Beatty and Shimshack, 2011; Moretti and Neidell, 2011; Janke, 2014; Schlenker and Walker, 2016; Simeonova et al., 2018; Alexander and Schwandt, 2019). Columns (1) and (2) in Table A.6 in the appendix show that LEZs in Germany have no statistically significant effect on respiratory disease.[18] The results on respiratory health are

[18] The ICD-10 codes for respiratory disease are J00–J06, J20–J22, J30–J47 and J95–J99.

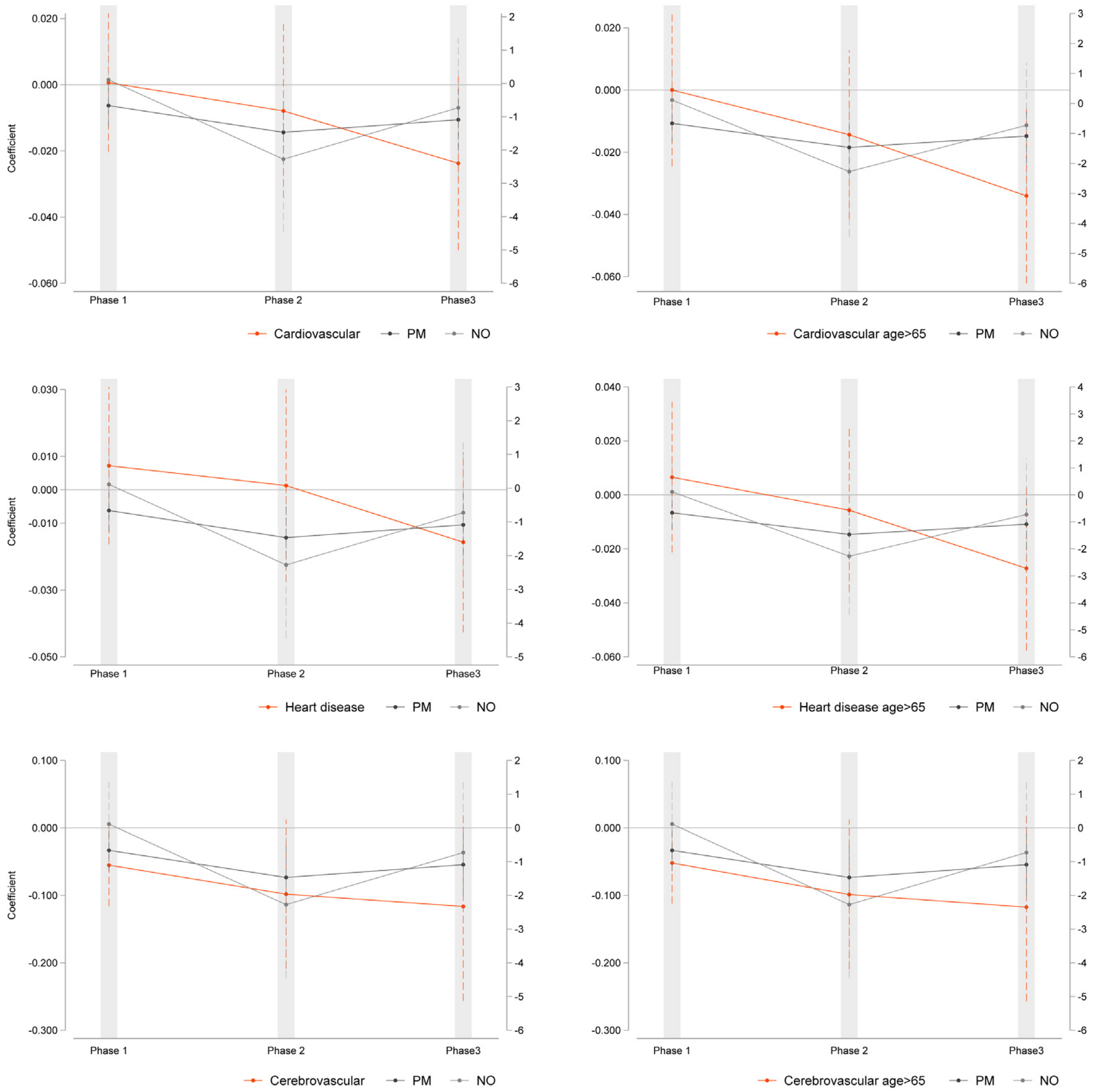

Fig. 9. LEZ phases and their impact on cardiovascular disease, PM_{10} and NO_2 separately. *Notes:* The figure presents coefficients from a subsample of treated cities that introduce LEZs after 2009. Coefficients and 95% confidence intervals come from equations akin to (1) and (3), where LEZ indicator is replaced with three indicators for each of the phases. Standard errors are clustered at the city level. *Data:* Zi, German Environmental Agency and German Weather Service.

somewhat at odds with the vast body of literature suggesting a strong link between pollution and respiratory health.[19] To be able to completely rule out any impact on respiratory health, it would be helpful to analyze medical prescription data as well. Nevertheless, this discrepancy may largely be explained by institutional specificity of Germany. In 2006, Germany has implemented a disease management program for chronic respiratory illnesses, particularly COPD and asthma. Patients are enrolled in these programs by their health practitioner, who decides on a regular and fixed check-up interval (either quarterly or half-yearly). Health insurance companies provide monetary incentives to physicians to include patients in these programs (Mehring et al., 2013; Steppuhn et al., 2016). If reductions in pollution are not large enough to revert or prevent disease, it might be statistically challenging to detect changes in the number of patients and visits in such an institutional environment. Multiple studies show that air pollution affects cognitive function and central nervous system, especially dementia (Power et al., 2016; Babadjouni et al., 2017; Calderón-Garcidue nas and Villarreal-Ríos, 2017; Bishop et al., 2018; Peters et al., 2019) and diabetes (Andersen et al., 2012; Thiering and Heinrich, 2015; Eze et al., 2015). Table A.6 shows the results for the diseases of the central nervous system, including dementia, in columns (3) and (4) and for diabetes in columns

[19] The respective result for children 0–6 is 0.010(0.017). Thus, there is no significant effect on children's respiratory health either.

Table 6
The effect of LEZ on cardiovascular disease. Hospital diagnosis statistics.

	Entire population (1)	Elderly above 65 (2)
Panel A: All cardiovascular disease		
LEZ	−0.037	−0.030
	(0.031)	(0.034)
Observations	726	726
Panel B: Heart disease		
LEZ	−0.048	−0.046
	(0.035)	(0.036)
Observations	726	726
Panel C: Cerebrovascular disease		
LEZ	−0.050	−0.042
	(0.052)	(0.058)
Observations	726	726
City FE	Yes	Yes
Year FE	Yes	Yes
Additional controls	Yes	Yes

Notes: The outcome is the number of patients with the given diagnosis per 10,000 population in logarithms. The additional controls are GDP per capita, unemployment rate, average age of the population and the number of deceased. Robust standard errors, clustered at the city level, in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Data:* RDC of the Federal Statistical Office and Statistical Offices of the Federal States, Hospital statistics, 2004–2014.

(5) and (6).[20] Across all specifications, the results suggest that the LEZ-induced reductions in air pollution do not have a large and statistically significant effect on these two outcomes. Disease management programs also exist for diabetes, hence, the lack of effect here as well might be specific to Germany's institutional setting.

5.5. *Additional results: hospital diagnosis statistics*

I supplement the evidence from outpatient data with inpatient data from the hospital admission records. The inpatient data offers the advantage of encompassing a longer time period before the rollout of the zones. To construct the outcome variable, I count the number of episodes with cardiovascular diagnosis for each calendar year for the entire population and for those over the age of 65 separately. To evaluate the effect of LEZs on health using hospital data, I estimate a variant of Eq. (3), where the outcome is the number of hospital admissions with the given diagnosis per 10,000 population in logarithms in city c in year t. I further add time-varying controls at the city level, that is, GDP per capita, unemployment rate, average age of the population and the number of deceased. Table 6 presents the results. All estimates point to reductions in hospital admissions with cardiovascular disease. However, none of the estimates is precisely estimated at the 95% significance level.[21]

Two shortcomings of hospital data might explain this imprecision. First, the data contains no information on whether the person lives inside or outside an LEZ. Second, the inpatient data includes the cases that end up in hospitals, hence while it is ideal for studying severe cases, it is less suited for studying cases that can be managed by outpatient care.

6. Identification threats and robustness checks

This section presents balancing regressions as a test of orthogonality and shows that the main results are robust to using a different control group, to sample restrictions, to alternative and placebo specifications, to different definitions of outcome and treatment variables.[22]

Balancing regressions. One potential threat to the identification strategy is that the implementation of LEZs might be correlated with simultaneous socioeconomic changes that affect pollution and health outcomes. To test for such violations, I run balancing regressions (Pei et al., 2019; Alexander and Schwandt, 2019). In the spirit of difference-in-differences strategy, I regress the socioeconomic characteristics of treated areas on the LEZ indicator, along with city and year fixed effects. For each city, I collect data on industrial output per capita, unemployment rate, GDP per capita, output of health and other services and population density for the years 2004–2016 from the Federal Statistical Office. Panel A in Table 7 shows the corresponding results. Reassuringly, the coefficients on the LEZ indicator are insignificant in all regressions.

In Panel B I obtain two additional datasets from the Research Data Centre Ruhr at the Leibniz-Institute for Economic Research (FDZ Ruhr). The first dataset – RWI-GEO-GRID – contains a range of socioeconomic variables collected by the commercial micro- and geo-marketing provider *microm*, which uses more than one billion individual data points for aggregation. The data is available for the year 2005 and 2009–2017. From this dataset, I use variables on purchasing power per household,[23] population density, and the share of adults of the age 30–65 and over the age of 65. The second dataset – RWI-GEO-RED (2020) – is a unique dataset on German real estate prices, obtained from the largest internet platform on real estate in Germany. The data covers the years 2007–2017. From this dataset I use offering monthly rental prices for flats.

Both the socioeconomic variables and the rental prices are available at the postal code level. For consistency, I aggregate them at the area level, as defined in Section 5. Columns (6) through (10) show that there is no evidence that household income, age composition, population density, and rental prices change systematically differently within LEZs.

Definition of the control group. One may worry that using entire cities as a control group for LEZs may have implications for the estimates. LEZs are primarily placed in the densely populated central areas of major cities. Comparing these areas to the entire area of untreated cities might result in biased estimates if city centers systematically follow different time trends in health outcomes.

[20] The respective ICD-10 codes are G00–G32, F00–09 and E10–E14.
[21] Fig. A.1 in Appendix A presents the event study graphs for the hospital diagnosis statistics.

[22] Table A.7 in the appendix presents the robustness checks for air pollution results.
[23] Purchasing power is calculated to reflect household income. It includes labor income, capital wealth, rental and leasing income minus taxes and social security contributions.

Table 7
Balancing regressions.

Panel A: Characteristics at city level					
	Industrial output per capita (1)	Unemployment rate (2)	GDP per capita (3)	Health and services per capita (4)	Population density (5)
LEZ	0.215	0.001	−447.2	−0.027	8.184
	(0.243)	(0.029)	(311.6)	(0.108)	(13.62)
Observations	897	897	897	897	897
City FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Panel B: Characteristics at LEZ level					
	Purchasing power (6)	Share of age > 65 (7)	Share of 30 < age < 65 (8)	Population density (9)	Rental prices (10)
LEZ	−110.627	−0.177	−0.091	−64.860	0.200
	(414.179)	(0.131)	(0.101)	(62.677)	(0.382)
Observations	1290	1290	1290	1290	719
Area FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variables in column headings. In Panel A, LEZ indicates whether a city has a low emission zone at a given time. In Panel B, LEZ indicates whether a given postal code is inside a low emission zone. Robust standard errors, clustered at the city level, in parentheses. *Data:* Federal Statistical Office of Germany (Panel A), RWI-GEO-GRID and RWI-GEO-RED (Panel B).

To address this concern, I generate hypothetical LEZs in untreated cities. Although municipal environmental agencies in each city follow an individual approach in placing their LEZs, one common predictor across all cities is the population density. The higher the density, the more likely it is that a postal code area is included in the LEZ. Hence, in untreated cities I draw the hypothetical LEZs by encircling the most densely populated postal code areas. For details on how the hypothetical LEZs are constructed, see Appendix B.

Table 8 presents the results from a regression akin to specification (3), however here the sample includes the existing and hypothetical LEZs only. The results support the main findings, suggesting that there is no evidence that city centers systematically follow different trajectories, despite differing in their socioeconomic characteristics at the baseline.

Specification issues. A potential concern is that the LEZs are a last resort policy for treated cities, which have enacted various anti-pollution policies previously. If those policies already beforehand put the treated cities on a different trajectory with respect to air pollution and health, the common trend assumption might be violated. However, the examination of the environmental plans of all cities included in the estimation sample shows that both treated and untreated cities have implemented various measures before the initial LEZs were enacted. These measures focused mainly on the expansion of public transport and cycling, retrofitting municipal vehicles, enacting parking policies and optimizing traffic flow. Since these policies were in place both in treated and untreated cities, they should affect all cities similarly. To further ensure that the results are not driven by differences in trends between treated and untreated cities, column (1) in Table 9 restricts the estimation sample to cities that have introduced an LEZ at some point during the observation period, meaning that the variation comes from the timing of the policy only. This robustness check also addresses the worry that treated and untreated cities are significantly different in their baseline characteristics, and hence might also systematically follow different trends. Reassuringly, the point estimates remain comparable to the main results.

Relatedly, the implementation of anti-pollution measures in untreated cities, if successful in reducing pollution, will lead to a downward bias in difference-in-differences estimates. Thus in column (2), I add a contemporaneous indicator variable for Clean Air Action Plans in untreated cities. The results show that the point estimates remain similar to the baseline results. Wolff (2014) also shows that there is little indication that other elements of Clean Air Action Plans have been effective.

Alternative outcomes. In the next robustness checks, I address the sensitivity of results with respect to the definition of the outcome. In column (3) of Table 9, the outcome is the number of cases instead of the number of patients, as in the main specification. This definition of the outcome should capture the intensive margin of the treatment effect as it also includes multiple visits by the same patient. The results suggest one percentage point larger drop in the number of patient visits versus the number of patients, suggesting that the reduction in the number of exacerbations plays a larger role.

Column (4) redefines the outcome in terms of prevalence rates, whereby prevalence rate is the fraction of patients with cardiovascular diagnosis in the total number of patients with any diagnosis in the respective age group. The results remain comparable in magnitude, however, the coefficient for the elderly sample becomes imprecisely estimated.

Placebo tests. The next block in Table 9 probes the sensitivity of results to three placebo tests. In column (5) I use the number of patients with injuries (ICD-10 W00-X59) as a placebo outcome. In column (6), I generate the timing of the LEZ enactments randomly for all treated cities. Subsequently, I drop all observations after the true treatment date and regress the outcome on the randomly generated LEZ indicator. In column (7), I drop all existing LEZs and use the number of patients with cardiovascular diagnosis

Table 8
The effect of LEZ on cardiovascular disease. Existing and hypothetical LEZs.

	Entire population		Elderly above 65	
	(1)	(2)	(3)	(4)
Panel A: All cardiovascular disease				
LEZ	−0.013	−0.015	−0.026*	−0.028*
	(0.017)	(0.018)	(0.015)	(0.016)
Observations	594	396	594	396
Panel B: Heart disease				
LEZ	−0.003	−0.008	−0.012	−0.016
	(0.023)	(0.024)	(0.022)	(0.022)
Observations	594	396	594	396
Panel C: Cerebrovascular disease				
LEZ	−0.048	−0.080**	−0.056*	−0.090***
	(0.029)	(0.030)	(0.029)	(0.029)
Observations	954	585	954	585
Area FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Intro after 2009	No	Yes	No	Yes

Notes: The outcome is number of patients with the given diagnosis in logarithms. Area FE are fixed effect referring either to existing or to hypothetical LEZs. The restriction "Intro after 2009" refers to keeping only those treated cities that introduced an LEZ after 2009. Robust standard errors, clustered at the city level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1.*Data:* Zi.

Table 9
Robustness checks.

	Specification issues		Alternative outcomes		Placebo tests		
	LEZ cities only (1)	Clean Air Action Plans (2)	Patient-cases (3)	Prevalence rate (4)	Pl. outcome: Injuries (5)	Pl. timing (6)	Hypothetical LEZs (7)
Entire population	−0.024*	−0.036***	−0.033***	−0.033**	0.372	0.000	−0.005
	(0.012)	(0.008)	(0.009)	(0.012)	(0.393)	(0.012)	(0.011)
Elderly above 65	−0.030**	−0.043***	−0.041***	−0.015	0.018	−0.004	−0.006
	(0.012)	(0.009)	(0.009)	(0.012)	(0.124)	(0.011)	(0.013)
Observations	360	585	585	585	585	333	603
Area FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Intro after 2009	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Column (1) excludes all cities with no LEZ by 2017. Column (2) additionally controls for Clean Air Action Plans with an indicator variable. Column (3) and (4) use alternative outcomes. In column (3) the outcome is the overall number of patient-cases in logs. In column (4) the outcome is the number of cardiovascular patients per 10,000 patients with any diagnosis, in logs. Column (5) uses the number of patients with injuries as the placebo outcome. Column (6) replaces the actual LEZ indicator with a randomly generated fake indicator. Column (7) uses hypothetical LEZs instead of existing LEZs. Robust standard errors, clustered at the city level, in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Data:* Zi.

in hypothetical LEZs in untreated cities as the outcome. Reassuringly, all three placebo tests return a small and insignificant coefficient.

Data aggregation at the city level. In the next block of robustness checks, I aggregate observations at the city level. This level of aggregation helps to examine whether the improvements in cardiovascular health are in the entirety of treated cities or only within the boundaries of LEZs. Panel A in Table 10 presents the results from a specification equivalent to the main regression, with the difference that outcomes are aggregated at the city level. The point estimates are very small and statistically insignificant both for the entire population and for the elderly. This emphasizes the importance of defining the treatment precisely.

To examine the issue further, panel B in Table 10 replaces the binary LEZ indicator with the size of the zone, calculated as a percentage of the LEZ in relation to the entire area of the city. This number is restricted between 0 and 1. The point estimate suggests that when the relative size of the zone increases by 10 percentage points, the number of patients with cardiovascular disease in treated cities decreases by 5% for the entire population and by 8% for those over the age of 65. Panel C further adjusts the outcome to the population size by calculating the number of patients per 10,000 inhabitants in the respective age group. The estimates suggest around 7% reduction in cardiovascular diagnoses per 10,000 inhabitants for a 10 percentage points increase in the relative size of the zone.

7. Conclusion

This paper evaluates the effect of LEZs on population health in Germany. Using the across space and over time variation in the implementation of LEZs, I first demonstrate that LEZs reduce monthly PM_{10} concentrations by 0.9 $\mu g/m^3$, which translates into a 3% decline. The findings for NO_2 suggest small and statistically insignificant reduc-

Table 10
Robustness checks: aggregation at the city level.

	Entire population (1)	Elderly above 65 (2)
Panel A: LEZ indicator for the city		
LEZ	−0.009	−0.013
	(0.014)	(0.015)
Observations	396	396
Panel B: Proportional size of the LEZs		
LEZ size	−0.049***	−0.054***
	(0.015)	(0.017)
Observations	396	396
Panel C: Number of patients per 10,000 inhabitants		
LEZ size	−0.077***	−0.050**
	(0.014)	(0.022)
Observations	396	396
City FE	Yes	Yes
Year fixed effects	Yes	Yes
Intro after 2009	Yes	Yes

Notes: In Panel A, LEZ indicator takes on the value one if the city has an LEZ, and zero otherwise. Panel B replaces the binary LEZ indicator with the size of the zone, calculated as a percentage of the LEZ in relation to the entire area of the city. Panel C further adjusts the outcome to the population size by calculating the number of patients per 10,000 inhabitants in the respective age group. Robust standard errors, clustered at the city level, in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Data:* Zi.

tions. Next, using novel outpatient health care data, I show that the zones improve cardiovascular health outcomes: they reduce the number of patients with cardiovascular disease by 2–3%. The effect is particularly strong for those over the age of 65. These results are robust to a range of robustness checks.

Despite the well-established understanding on the adverse effects of pollution, little empirical evidence exists on the instruments targeting air quality and their impact on health. The paper adds to this growing literature by providing evidence on the effect of LEZs. Furthermore, the findings in this paper show that adult population, especially those over the age of 65, being susceptible to cardiovascular health conditions, are an important group to study in the context of evaluating the effects of pollution on health. Finally, the paper adds to the small number of studies analyzing the effects of long-term marginal reductions in pollution in a setting with relatively low pollution levels and shows that, even at comparably low levels of air pollution, improvements in air quality generate health benefits.

The empirical findings in this paper have strong policy implications. They demonstrate that LEZs are a helpful tool to reduce air pollution in urban areas and to improve health outcomes commonly related to air pollution. A back-of-the-envelope cost-benefit analysis suggests health benefits of nearly 4.43 billion Euro that have come at a cost of 2.3 billion Euro for vehicle upgrading. However, the distributional effects of the policy are less clear. Since the policy mainly targets high-emitting cars, which tend to be old and cheap cars, the practical burden of LEZs most likely falls on families from low socioeconomic background and on small businesses. In the meantime, the car fleet has changed dramatically since the early introduction days of the zones. In 2019, the share of cars that received a green badge was above 90%. This weakens the impact of LEZs. Furthermore, the analysis in this paper indicates that the improvements in air quality, while meaningful, are not large enough to revert the existing health trajectories. If cities aim at reducing the air pollution further, stricter policy measures are necessary.

Appendix A. Additional figures and tables

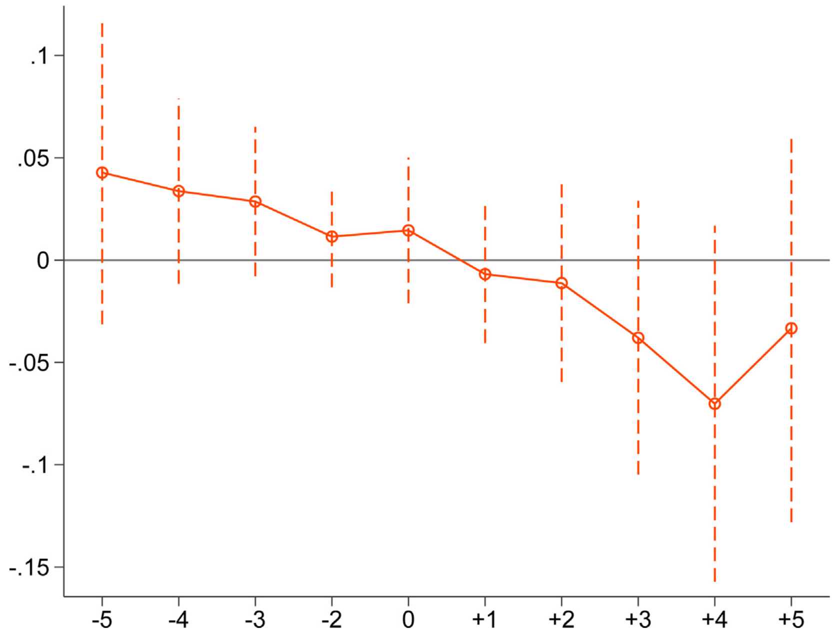

Fig. A.1. Event study of cardiovascular diseases. Hospital data. *Notes:* The outcome variable is the logarithm of the number of patients with cardiovascular disease per 10,000 population. Coefficients and 95% confidence intervals. Robust standard errors, clustered at the city level, in parentheses. *Data*: RDC of the Federal Statistical Office and Statistical Offices of the Federal States, Hospital statistics, 2004–2014.

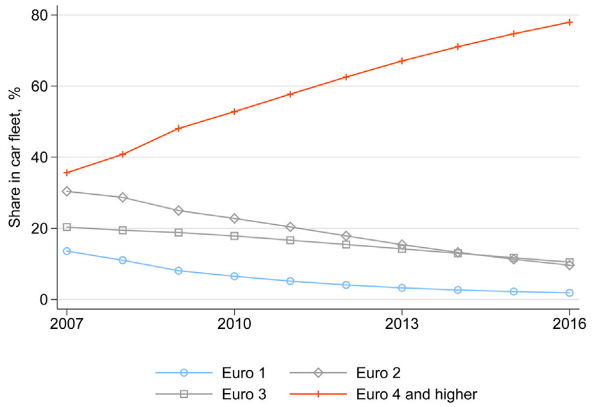

Fig. A.2. Development of passenger vehicle registration by emission class. *Notes*: Share of each Euro-class is calculated as its percentage in the total registration of cars annually in a city. *Data:* Federal Motor Vehicle Authority.

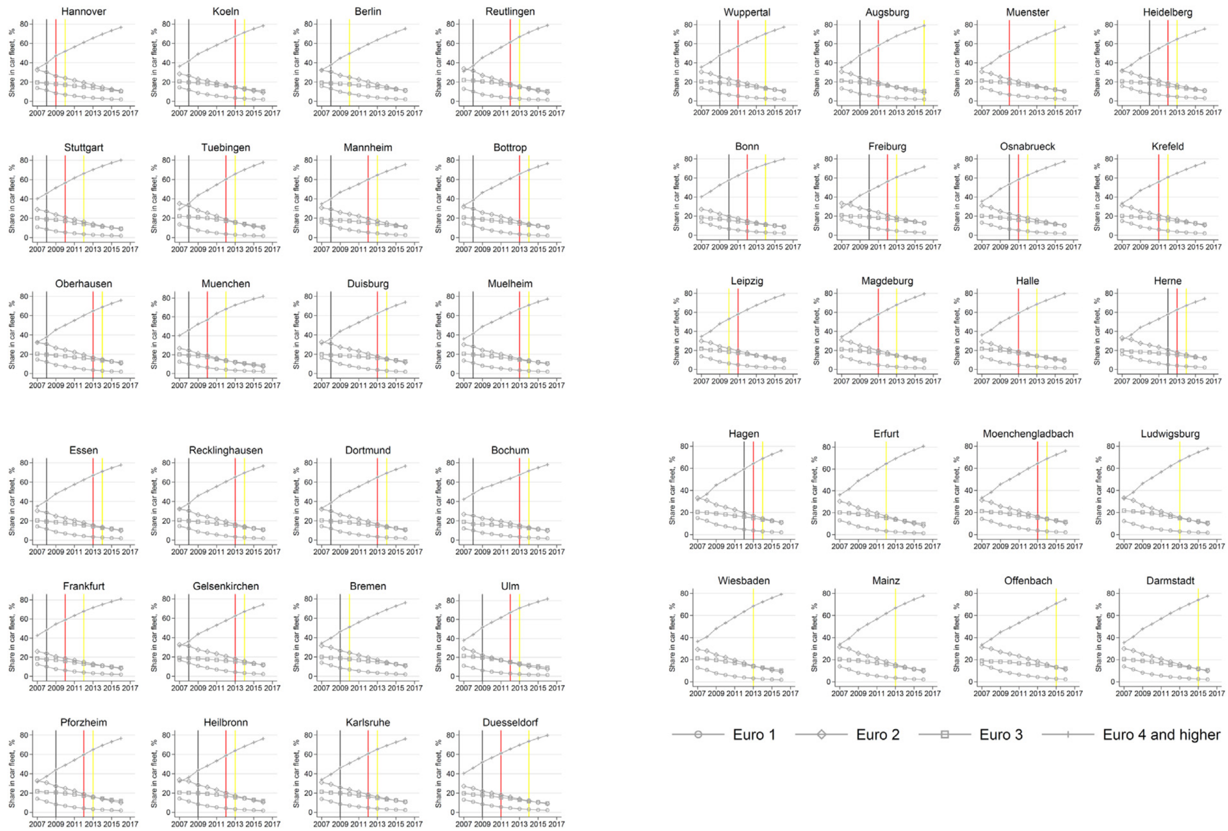

Fig. A.3. Development of passenger vehicle registration by emission class in treated cities. *Notes:* The figures plot the share of cars in each emission class for each treated city separately. The gray horizontal line signifies Phase 1, the red line signifies Phase 2 and the yellow line-Phase 3. *Data*: Federal Motor Transport Authority.

Table A.1
Emission classes, color codes and phase restrictions.

Emission class	Color code	Banned in	Description
Euro 4	Green	None	Petrol: CO: 1.00 g/km HC: 0.10 g/km NOx: 0.08 g/km Diesel: CO: 0.50 g/km HC + NOx: 0.3 g/km PM: 0.025 g/km
Euro 3	Yellow	Phase 3	Petrol: CO: 2.30 g/km HC: 0.20 g/km NOx: 0.15 g/km Diesel: CO: 0.64 g/km HC: 0.56 g/km NOx: 0.50 g/km PM: 0.05 g/km
Euro 2	Red	Phase 2	Petrol: CO: 2.20 g/km HC + NOx: 0.50 g/km Diesel: CO: 1.00 g/km HC + NOx: 0.70 g/km PM: 0.08 g/km
Euro 1	None	Phase 1	Petrol CO: 2.72 g/km HC + NOx: 0.97 g/km Diesel: CO: 2.72 g/km HC + NOx: 0.97 g/km PM: 0.14 g/km

Notes: The table displays European emission standards for exhaust emissions of new vehicles.

Table A.2
Typical cars in Euro-class emission category classification.

Euro 1	Euro 2	Euro 3	Euro 4 and higher
Volkswagen Caddy	Volkswagen Passat Variant	Nissan Primera	Volkswagen Tiguan
Mercedes-Benz E250	Mercedes-Benz C220	Mercedes-Benz Vaneo	Skoda Kodiaq
Audi 80 TDI	Citroen Xantia	Volvo V70	Mitsubishi Pajero
Volkswagen T4 California	Mercedes-Benz ML 270	BMW 520	Ford Fiesta
Volvo 940	BMW 740	Fiat Scudo	Ford Transit
		BMW 330	Volvo XC 60
		Opel Vivaro	Tesla
			Civic hybrid
			Prius

Notes: The table displays the typical cars in Euro classifications.

Table A.3
Introduction date and the areal coverage of LEZs.

City	Introduction	Coverage in %	Attainment Status
Augsburg	July 2009	3.95	Non-attainment
Berlin	January 2008	9.88	Non-attainment
Bochum[a]	October 2008	39.85	
Bonn	January 2010	6.38	Attainment
Bottrop[a]	October 2008	24.85	Non-attainment
Bremen	January 2009	2.18	
Cologne	January 2008	7.42	Attainment
Darmstadt	November 2015	87.64	Non-attainment
Dortmund[a]	October 2008	6.77	Non-attainment
Duisburg[a]	October 2008	18.47	Non-attainment
Dusseldorf	February 2009	19.78	
Erfurt	October 2012	5.86	Non-attainment
Essen[a]	October 2008	66.56	Non-attainment
Frankfurt	October 2008	44.3	Non-attainment
Freiburg	January 2010	16.18	Non-attainment
Gelsenkirchen[a]	October 2008	19.06	Non-attainment
Hagen	January 2012	5.39	Non-attainment
Halle	October 2011	5.1	Non-attainment
Hannover	January 2008	21.05	Non-attainment
Heidelberg	January 2010	9.29	Attainment
Heilbronn	January 2009	38.34	Non-attainment
Herne[a]	January 2012	100	
Karlsruhe	January 2009	6.52	Non-attainment
Krefeld	January 2011	23.2	Non-attainment
Leipzig	March 2011	61.35	Non-attainment
Ludwigsburg	January 2013	100	Non-attainment
Magdeburg	October 2011	3.33	Non-attainment
Mainz	February 2013	34.95	Non-attainment
Mannheim	March 2008	4.67	Non-attainment
Mönchengladbach	January 2013	12.38	Attainment
Mülheim[a]	October 2008		Attainment
Munich	October 2008	14.16	
Münster	January 2010	0.47	Attainment
Oberhausen[a]	October 2008	30.87	
Offenbach	January 2015	85.77	

Table A.3 (*Continued*)

City	Introduction	Coverage in %	Attainment Status
Osnabrück	January 2010	14.11	Attainment
Pforzheim	January 2009	1.99	Non-attainment
Recklinghausen[a]	October 2008	30.08	
Reutlingen	March 2008	100	Non-attainment
Stuttgart	March 2008	98.44	Non-attainment
Ulm	January 2009	23.07	Non-attainment
Wiesbaden	February 2013	31.12	Attainment
Wuppertal	February 2009	14.61	Non-attainment

Notes: Dates of introduction of the zones come from the *Umweltbundesamt*. The coverage refers to the relative size of LEZ in relation to the area of the city. The size of LEZ has been calculated based on shapefiles from *OpenStreetMap*.

[a] The cities in Ruhr area united into a common LEZ in January 2012.

Table A.4
Differences between LEZ and non-LEZ areas in treated cities in 2005.

	Purchasing power	Population density	Car density	Unemployment rate	Share of 30–65	Share of 65+
Mean treated	37,359	4865	0.72	10.28	49.87	17.73
Mean untreated	39,671	1899	0.94	8.40	49.44	19.45
Difference	−2311	2966	−0.217	1.888	0.429	−1.723
p-value	0.026	0.000	0.000	0.018	0.188	0.001

Notes: N treated = 43 and N control = 38. Purchasing power and car density are calculated per household. Populations density refers to number of people per km^2. *Data*: RWI-GEO-GRID.

Table A.5
The effect of LEZ on monthly $PM_{2.5}$ concentrations.

	(1)	(2)	(3)
LEZ	−0.435**	−0.332	−0.472*
	(0.205)	(0.210)	(0.253)
Observations	7114	6922	6594
Station fixed effects	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes
Weather controls	No	Yes	Yes
Station restrictions	No	Yes	Yes

Notes: The outcome is the monthly concentration of $PM_{2.5}$. Column (1) includes all stations and does not control for weather covariates. Column (2) includes all stations and controls for weather covariates. Column (3) restricts the sample to stations that functioned at least six months before and after the introduction of the respective LEZ zone. Robust standard errors, clustered at the city level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Data:* German Environment Agency and German Weather Service.

Table A.6
The effect of LEZ on further outcomes.

	Respiratory disease		CNS disease		Diabetes	
	All ages (1)	Over 65 (2)	All ages (3)	Over 65 (4)	All ages (5)	Over 65 (6)
LEZ	0.004	−0.065	−0.005	0.003	−0.004	−0.011
	(0.006)	(0.060)	(0.014)	(0.041)	(0.007)	(0.010)
Observations	585	585	585	585	585	585
Area FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Intro after 2009	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The outcome is number of patients with the given diagnosis in logarithms. Area FE refers to fixed effect for each district over which the numbers are aggregated. The restriction “Intro after 2009” refers to keeping only those treated cities that introduced an LEZ after 2009. Robust standard errors, clustered at the city level, in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Data:* Zi.

Table A.7
The effect of LEZ on monthly PM_{10} and NO_2 concentrations. Robustness checks.

	Monthly PM_{10}		Monthly NO_2	
	(1)	(2)	(3)	(4)
Panel A: Only LEZ cities				
LEZ	−0.635**	−0.656*	−0.581	−1.014***
	(0.273)	(0.360)	(0.371)	(0.348)
Observations	13,014	13,014	13,012	13,012
Panel B: Restricted observation period				
B.1: Excluding 2004 and 2005				
LEZ	−0.796***	−0.749**	−0.562*	−1.053**
	(0.260)	(0.301)	(0.304)	(0.437)
Observations	16,477	16,477	17,293	17,293
B.2: Excluding 2013 and 2014				
LEZ	−0.693**	−0.859**	−0.389	−0.795**
	(0.322)	(0.328)	(0.368)	(0.354)
Observations	16,357	18,099	17,039	17,039
B.3: Excluding 2004–2005 and 2013–2014				
LEZ	−0.705**	−0.823**	−0.778**	−1.259***
	(0.308)	(0.340)	(0.296)	(0.345)
Observations	13,025	13,025	13,528	13,528
Panel C: Placebo timing				
LEZ	0.363	0.132	0.327	0.406
	(0.372)	(0.381)	(0.601)	(0.696)
Observations	13,562	13,562	14,542	14,542
Station fixed effects	Yes	Yes	Yes	Yes
Year-month fixed effects	Yes	Yes	Yes	Yes
Weather controls	Yes	Yes	Yes	Yes
Station restrictions	No	Yes	No	Yes

Notes: The outcome is the monthly concentration of PM_{10} and NO_2. Panel A restricts the sample to treated cities only. Panel B restricts the observation period. Panel C regressed the outcome on a randomly generated LEZ introduction date. Robust standard errors, clustered at the city level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Data*: German Environment Agency and German Weather Service.

Appendix B. Construction of hypothetical LEZs

Municipal environmental agencies in each city follow an individual approach in placing their LEZs. This generates a vast variation in the characteristics and proportional size of LEZs across treated cities. Nevertheless, across treated cities population density in a postal code is a common predictor whether that postal code is inside an LEZ. The higher the density, the more likely it is that a postal code area is included in the LEZ. Columns (1) and (2) in Table B.1 show the strong and significant correlation between the probability that a postal code area is part of an LEZ and population density. A descriptive analysis also shows that treated postcode areas are at least in the top 50th percentile of population density distribution. Furthermore, all LEZs, at least partially, cover the center of the city.

To generate the hypothetical LEZs, in analogy to treated cities, I define the postal code areas that fall in the top 50th percentile according to population density in untreated cities. I define the density cutoffs for each city separately, instead of defining an absolute cutoff across the entire sample. Since there is a large between-city variation, an absolute density cutoff would lead to no potential postal codes for hypothetical LEZ construction in certain cities. On a choropleth map with postal codes shaded in proportion to population density, I manually define the hypothetical LEZs for untreated cities, encircling postal codes in the top 50th percentile of density distribution that fall in the city center. To ensure a contiguous region in the center of the city, I exclude the postal code areas that are not located in the city center, even when these fall in the top percentiles

Table B.1
LEZs and population density at the postal code level.

	Existing LEZ		Hypothetical LEZ	
	(1)	(2)	(3)	(4)
Density	0.039***	0.055***	0.074***	0.099***
	(0.005)	(0.004)	(0.011)	(0.016)
Observations	923	923	345	345
R-squared	0.109	0.436	0.262	0.481
City FE	No	Yes	No	Yes

Notes: The outcome is binary and equals one if the postal code is inside an existing or hypothetical LEZ. Robust standard errors, clustered at the city level, in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. *Data:* Census 2011.

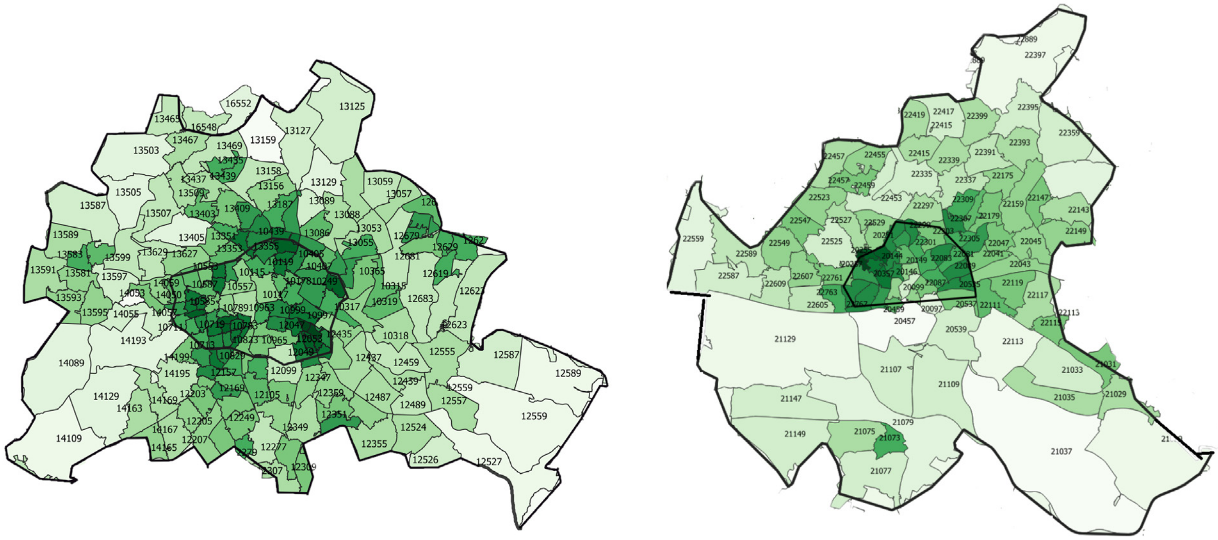

Fig. B.1. Population density by postal codes in Berlin and Hamburg. *Notes:* The figures show the population density per sq/km. The darker the shade of the green, the more densely is the postal code populated. On the left is Berlin, with its existing LEZ contoured. On the right is Hamburg, with its hypothetical LEZ contoured.

of density distribution.[24] Fig. B.1 also illustrates this exercise visually on the example of the map of Berlin (treated) and Hamburg (untreated). Columns (3) and (4) in Table B.1 show the correlation between the probability that a postal code is included in the hypothetical LEZ and population density. As before, the strong and significant correlation remains.

Appendix C. Cost-benefit analysis

An official cost-benefit assessment, unfortunately, is not available. This section presents a cost-benefit assessment under multiple subjective assumptions. This analysis is intended to provide a rough direction rather than a precise assessment.

Calculating the cost of LEZs requires multiple arbitrary assumptions. Although various factors are part of the total cost, clearly, the largest costs are due to vehicle upgrade. Here the difficulty is that it is unclear which part of this upgrade would have happened in the absence of LEZs and which is directly the result of LEZs.

I calculate the costs of car fleet update (direct costs) following the strategy in Wolff (2014). Following the author, I assume that the average cost of upgrading a car is 1650 Euro. First, I calculate the yearly percentage change in green sticker cars in treated cities. I then subtract the lowest average pace of fleet update in control cities (2.5%) to approximate the change in car fleet that is independent of the LEZs. I multiply this final percentage with the total number of cars and multiply this number by 1650. Finally, I sum up the costs over all treated cities in their respective post-treatment period. This results in total costs of 2.3 billion Euro. It is worth noting that there are other cost factors that this number does not take into account, such as time lost due to taking longer routes or using other means of transportation (public transport, car sharing, cycling) and the potential negative effect of LEZs on (small) businesses inside the zone.

Calculating benefits generated by the LEZs is complicated as well since there is no official, publicly available cost estimate by patient age groups and diagnoses. Moreover, proper benefit calculations would require factoring in not only direct, but also indirect benefits, such as the utility to residents from low pollution. Keeping all shortcomings in mind, below I follow a simple approach in calculating the benefits generated by fewer number of cardiovascular diagnoses.

According to the cost of illness report of the German Statistical Office (Statistisches Bundesamt, 2015), per capita cost of illness of cardiovascular disease is 570 Euro. Meanwhile, the per capita cost of cardiovascular disease for those over the age of 65 is 2600 Euro. General cost breakdown suggests that around 50% of all healthcare costs are generated by the outpatient care. This implies that ambulatory healthcare per capita cost is 235 Euro for overall population and 1300 Euro for those over the age of 65. The Robert Koch Institute reports that 8.5% of overall population and 23% of those older than 65 are diagnosed with a cardiovascular disease. I use the latter statistic to calculate the costs per patient for general population and for the elderly. The calculation leads to 2764.7 Euro savings for overall population (100/8.5*235) and 5652.2 Euro for the elderly (100/23*1300):

My estimates in Section 5 suggest that LEZs reduced the number of patients with a cardiovascular diagnosis by 3.3%. Using these numbers, I calculate that the outpatient health

[24] An example of such a postal code area is postal code 21073 in Hamburg in Figure B.1.

care system saved 4.43 billion Euro.[25] The cost-benefit calculation in this section suggests that the health benefits generated by LEZs outweigh the costs by 2.2 billion Euros.

Appendix D. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.jhealeco.2020.102402.

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[25] This number, however, does not take into account the costs related to forgone working time for previously employed people. These costs according to a recent study by Weimar et al. (2003) average at 18,429 Euro per employed patients.

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